

# EE204 Introduction to Semiconductor Devices

## **MOS Capacitor & MOSFET**

# Outline (1/3)

- **MOS Capacitor**
  - Basic structure
  - Biasing modes
- **Ideal MOS Capacitor**
- **Energy Band Diagram**
- **Solving Poisson's Equation**
  - Depletion approximation

# Outline (2/3)

- **Threshold Voltage**

- Physical definition
- Mathematical expression

- **MOS Capacitance**

- Equivalent electrical circuit
- Capacitance-voltage curves
- High and low frequency capacitance
- Deep depletion

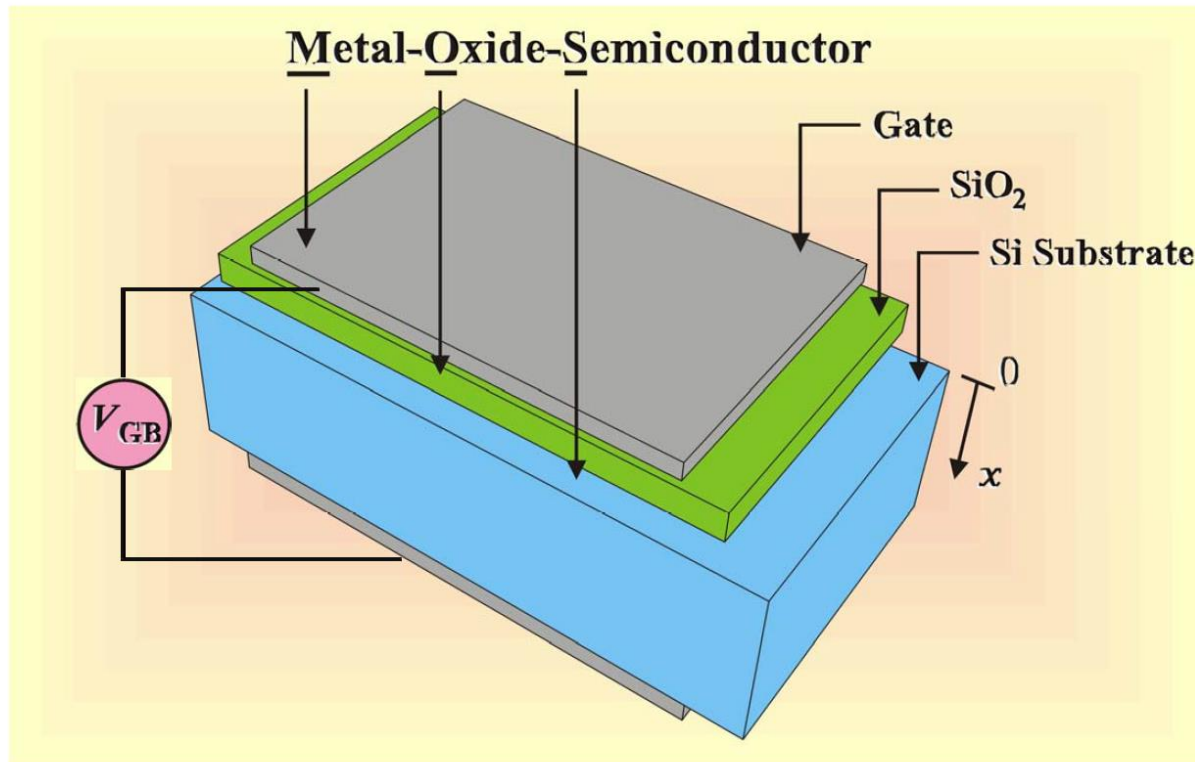
# Outline (3/3)

- **Non-ideal MOS Capacitor**
  - Work function difference
  - Oxide and interface trapped charge
  - Poly-silicon depletion
- **Flat band Voltage**
  - Concept and definition
  - Mathematical expression
- **MOSFET**

# MOS Capacitor

- A two-terminal device.
  - A thin silicon dioxide ( $\text{SiO}_2$ ) layer, “sandwiched” between a metal gate and a silicon (Si) substrate.
- A common test structure for extracting important device parameters:
  - Oxide thickness
  - Oxide charge density
  - Substrate doping concentration

# MOS Capacitor Structure



# MOS Capacitor

- A standard test vehicle for evaluating oxide reliability.
- A building block of other MOS devices:
  - MOSFETs (Metal-Oxide-Semiconductor Field-Effect Transistors)
  - CCDs (Charge-Coupled Develops)

# Ideal MOS Structure (1/2)

- **Properties:**

- *Zero work function difference*

- Gate and Si substrate have **equal** work functions.

- *Perfect insulator layer*

- No current flows under all *static* biasing conditions.
    - Zero defect density - no charges in the oxide and at the oxide-semiconductor interface.

- *Ohmic back contact*

- Majority carriers are supplied or removed spontaneously through the back contact.



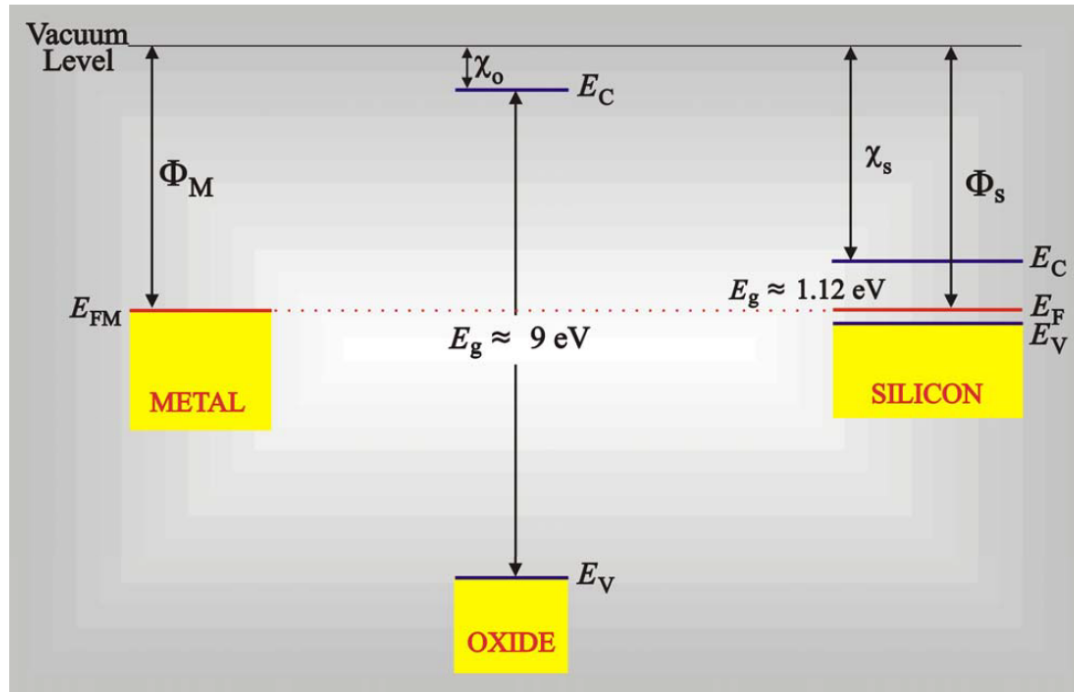
# Ideal MOS Structure (2/2)

- **Properties** (cont'd):

- *Infinitely thick Si substrate*

- The thickness of the Si substrate is much larger than the maximum width of the depletion region.
    - A neutral region, commonly known as the Si “bulk”, thus exists near the back contact under all biasing conditions.

# Energy Band Diagram



Individual energy band diagrams of the metal,  $\text{SiO}_2$ , and Si substrate of the MOS structure.

# Energy Band Diagram

- **Symbols:**

- $\Phi_M$  = work function of metal gate
- $\Phi_S$  = work function of Si
- $\chi_o$  = electron affinity of  $\text{SiO}_2$
- $\chi_s$  = electron affinity of Si
- $E_g$  = band gap energy
- $E_C$  = conduction band edge
- $E_V$  = valence band edge
- $E_{FM}$  = Fermi level in metal gate
- $E_F$  = Fermi level in Si

# Energy Band Diagram

- **Question:**

- Write down an expression for the work function of the p-type Si substrate.

**Recall:**

$$N_A \approx p = N_V \exp[(E_V - E_F)/kT]$$
$$E_V - E_F = kT \ln(N_A / N_V)$$

- **Answer:**

- Work function of Si:

$$\begin{aligned}\Phi_s &= \chi_s + (E_C - E_F) \\ &= \chi_s + [E_g - (E_F - E_V)] \\ &= \chi_s + [E_g + kT \ln(N_A / N_V)]\end{aligned}$$

$$N_C = 2.8 \times 10^{19} \text{ cm}^{-3}$$

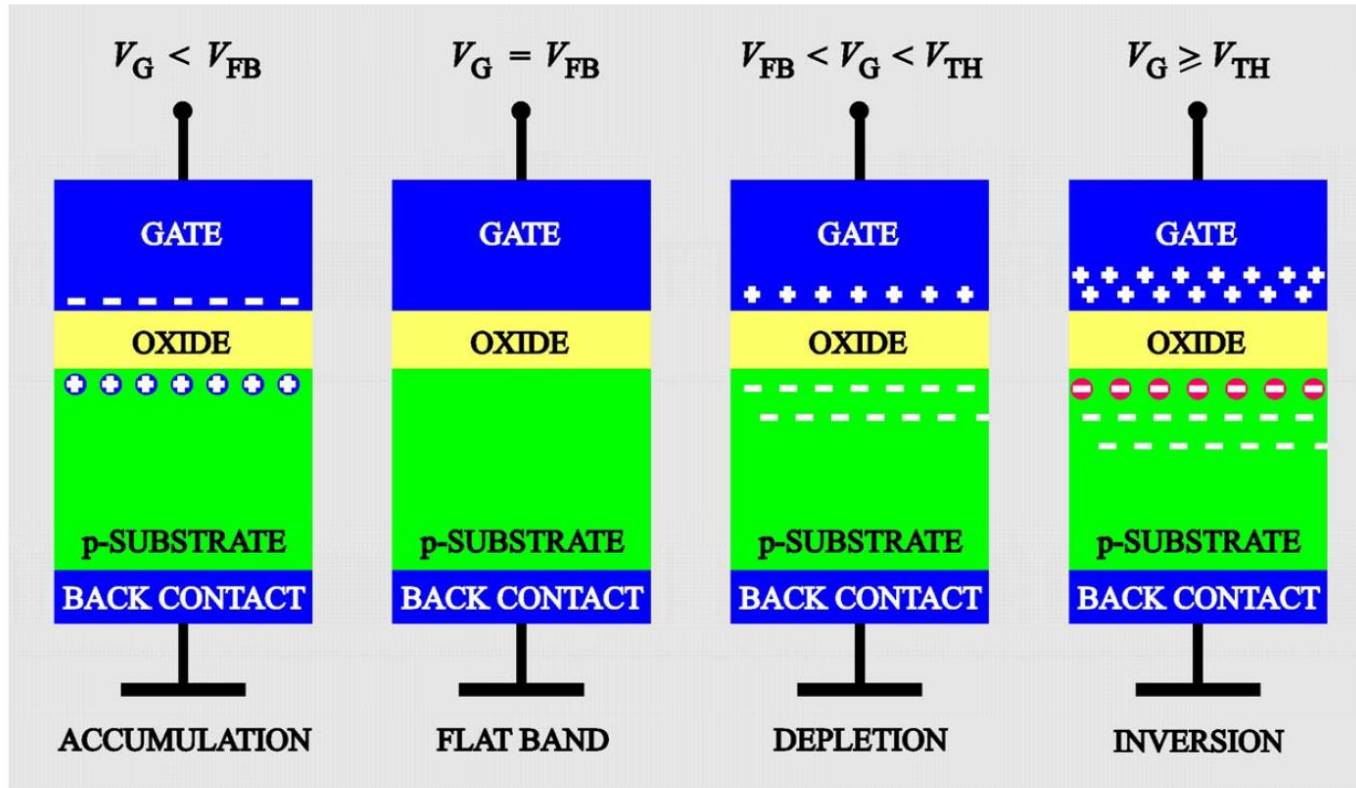
$$N_V = 1.04 \times 10^{19} \text{ cm}^{-3}$$

# Energy Band Diagram

- **Question:**

- An MOS capacitor is fabricated on a p-type Si substrate, with a uniform doping concentration of  $10^{17}\text{cm}^{-3}$ .
  - Calculate the work function of the Si substrate. The electron affinity of Si is 4.05eV.
  - What should be the doping type and concentration in order for the Si substrate to have the same work function as the gate? The work function of the gate is 4.25eV.

# Biasing Modes

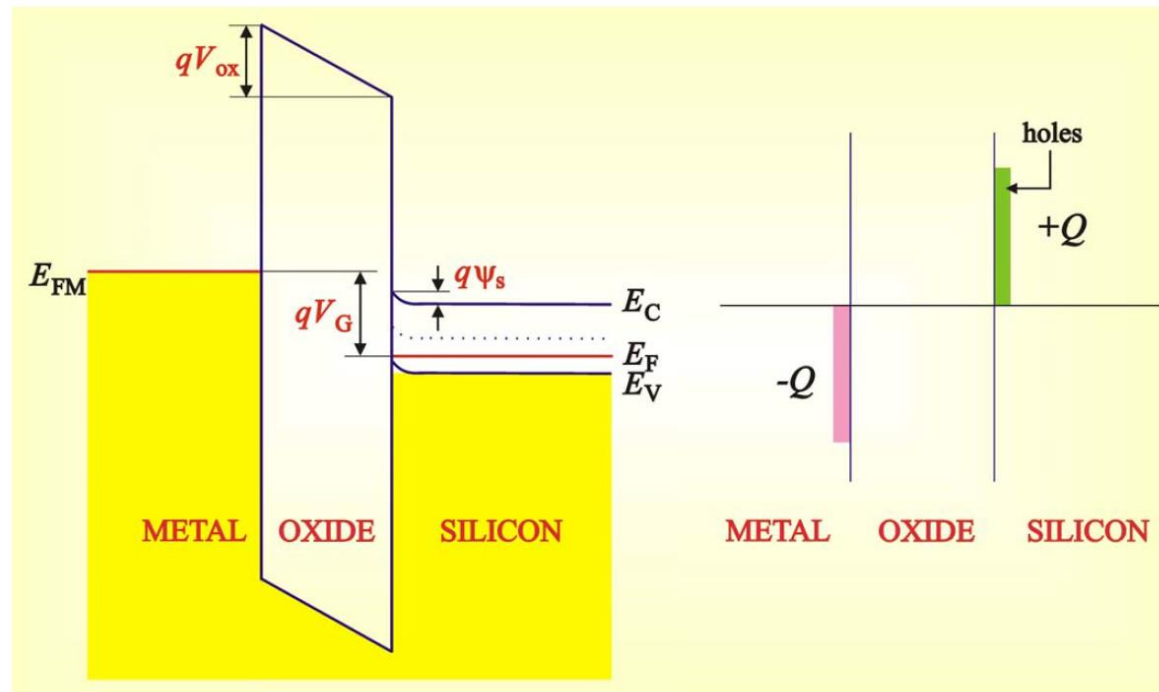


# Biasing Modes

- **Accumulation**

- Gate voltage is negative (positive) with respect to the p-type (n-type) Si substrate.
- Negative (positive) field attracts holes (electrons) to the surface region, i.e. the concentration of majority carriers near the surface increases.
- A pile-up of majority carriers at the surface is referred to as accumulation.

# Energy Band Diagram - Accumulation



Left: Energy band diagram of an ideal NMOS capacitor in accumulation

Right: Corresponding block charge diagram



# Energy Band Diagram - Accumulation

- Because there is no dc current flow (due to the insulating oxide layer), both  $E_{\text{FM}}$  and  $E_{\text{F}}$  are independent of position.
- The difference between  $E_{\text{FM}}$  and  $E_{\text{F}}$  gives the applied gate voltage in eV, i.e.  $qV_{\text{G}}$ .
  - As the applied gate voltage must be negative in order to induce an accumulation layer in the NMOS capacitor,  $E_{\text{FM}}$  “moves upwards” relative to  $E_{\text{F}}$  by an amount of  $qV_{\text{G}}$ .

# Energy Band Diagram - Accumulation

- The increase in hole concentration at the surface is denoted by a smaller difference between  $E_F$  and  $E_V$  at the surface than in the Si bulk.
- An energy barrier of 4.7eV between the  $E_V$  of Si and that of the  $\text{SiO}_2$  prevents the holes from flowing across the oxide layer.
- The corresponding energy barrier for electrons is 3.1eV for the Si- $\text{SiO}_2$  interface.

# Accumulation

- **Question:**

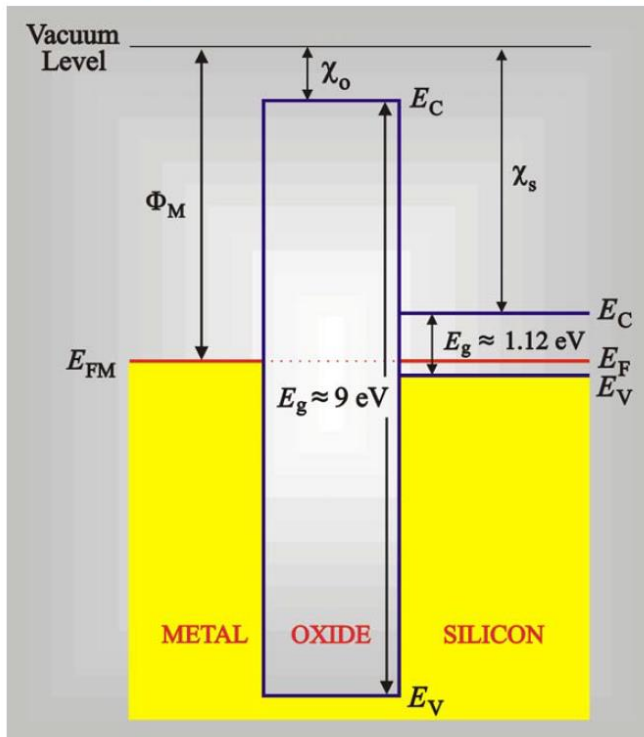
- An ideal MOS capacitor is fabricated on a uniformly doped p-Si substrate. The doping concentration is  $5 \times 10^{15} \text{ cm}^{-3}$ . A negative voltage of  $-3\text{V}$  is applied to the gate.
  - What is the voltage drop across the oxide layer, if it is known that the surface concentration of the accumulation layer is  $5 \times 10^{17} \text{ cm}^{-3}$ ?
  - Draw the energy band diagram of the MOS capacitor.

# Biasing Modes

- **Flat band**

- The corresponding gate voltage is referred to as the flat band voltage  $V_{\text{FB}}$ .
- For an ideal MOS structure,  $V_{\text{FB}} = 0$ .
- The Si substrate is neutral under flat band condition.

# Energy Band Diagram



The Fermi levels in the metal and Si are aligned under thermal equilibrium condition, i.e.  $V_G = 0$ .

For an ideal MOS capacitor,  $V_G = 0$  corresponds to flat band condition in the Si substrate.

Energy band diagram of an ideal MOS structure, upon contact of the metal gate,  $\text{SiO}_2$  and Si substrate.

# Biasing Modes

- **Depletion** (1/2)

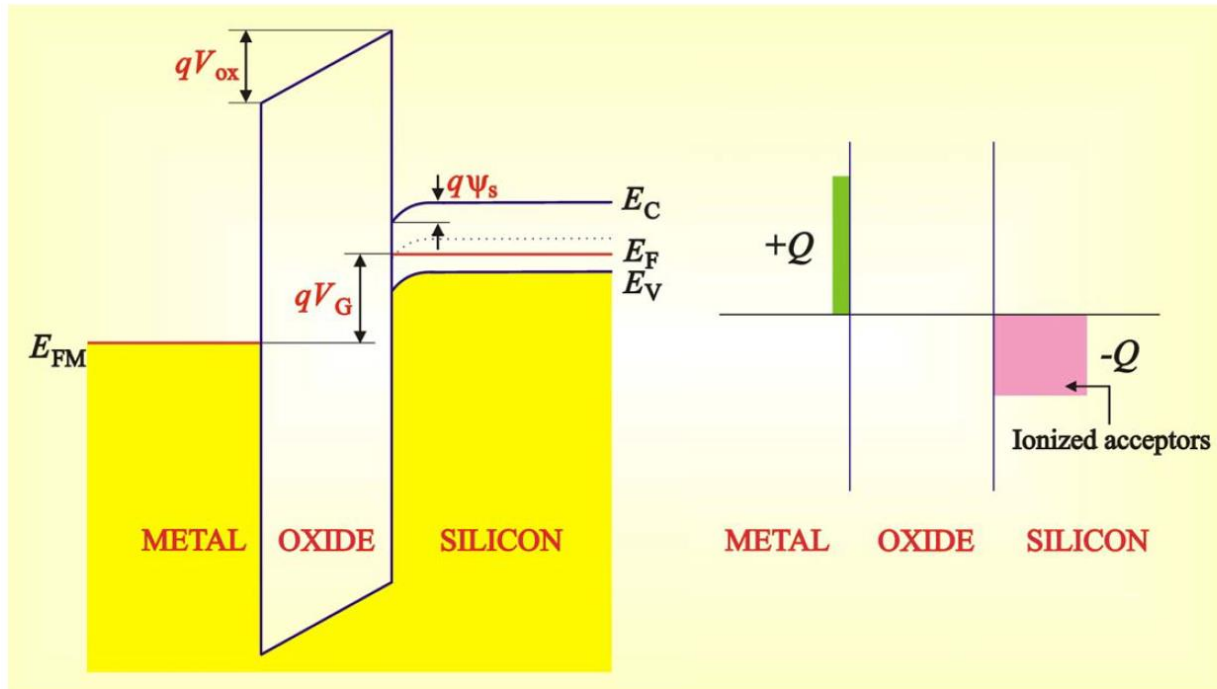
- Gate voltage is slightly positive (negative) with respect to the p-type (n-type) Si substrate.
- Positive (negative) field repels holes (electrons) away from the surface region, i.e. the concentration of majority carriers at the surface decreases.
- The departure of majority carriers leaves behind immobile acceptor (donor) ions in the p-type (n-type) Si substrate.

# Biasing Modes

- **Depletion** (2/2)

- The MOS capacitor is said to have entered into depletion.
- Note, however, that minority carriers are attracted to the surface. Hence, their concentration at the surface increases slightly.

# Energy Band Diagram - Depletion



Left : Energy band diagram of an ideal NMOS capacitor in depletion

Right : Corresponding block charge diagram

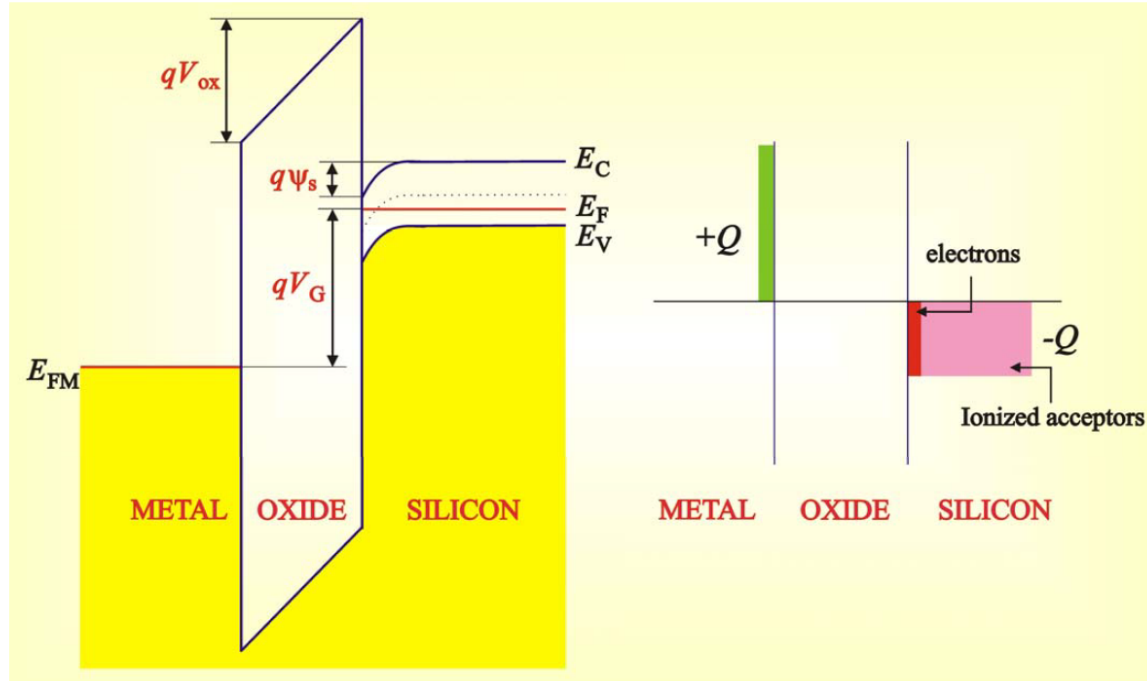


# Biasing Modes

- **Inversion**

- The concentration of minority carriers at the surface increases as the gate voltage becomes increasingly more positive (negative) with respect to the p-type (n-type) Si substrate.
- When the surface concentration of minority carriers is equal to the concentration of majority carriers in the neutral Si substrate, the surface is said to have been “inverted”.
- The corresponding gate voltage is called the threshold voltage  $V_{TH}$ .

# Energy Band Diagram – Onset of Inversion



Left : Energy band diagram of an ideal NMOS capacitor at onset of inversion

Right : Corresponding block charge diagram

# Onset of Inversion

- Physical definition of inversion:
  - The minority carrier concentration at the surface is equal to the majority carrier concentration in the Si bulk, i.e.

$$E_F - E_i(x=0) = E_i(x \rightarrow \infty) - E_F$$

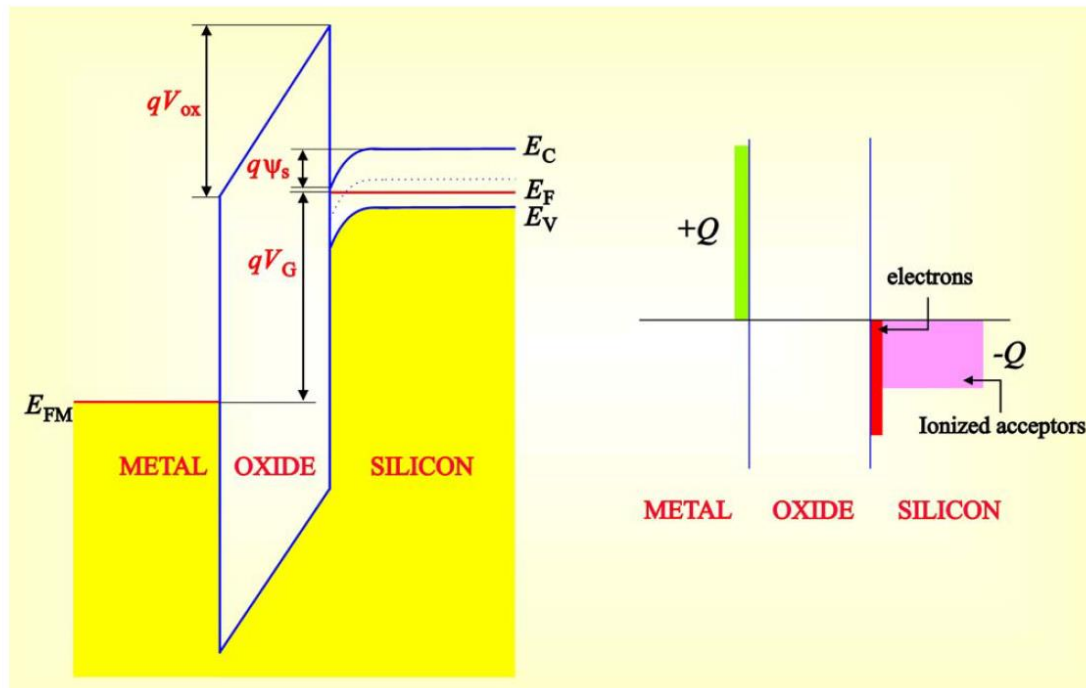
- Band bending:

$$q\psi_s = 2[E_i(x \rightarrow \infty) - E_F]$$

# Strong Inversion

- After the onset of inversion, a further increase in  $V_G$  places the MOS capacitor in strong inversion.
- Salient features:
  - Band bending  $q\psi_s$  increases only slightly (several tens of mV), due to the exponential dependence of surface carrier concentration on  $q\psi_s$ .
  - Depletion width practically remained constant. The increase in positive gate charge is almost entirely compensated by the increase in inversion charge.

# Energy Band Diagram – Strong Inversion



Left : Energy band diagram of an ideal NMOS capacitor in strong inversion

Right : Corresponding block charge diagram

# Poisson's Equation

- General expression:

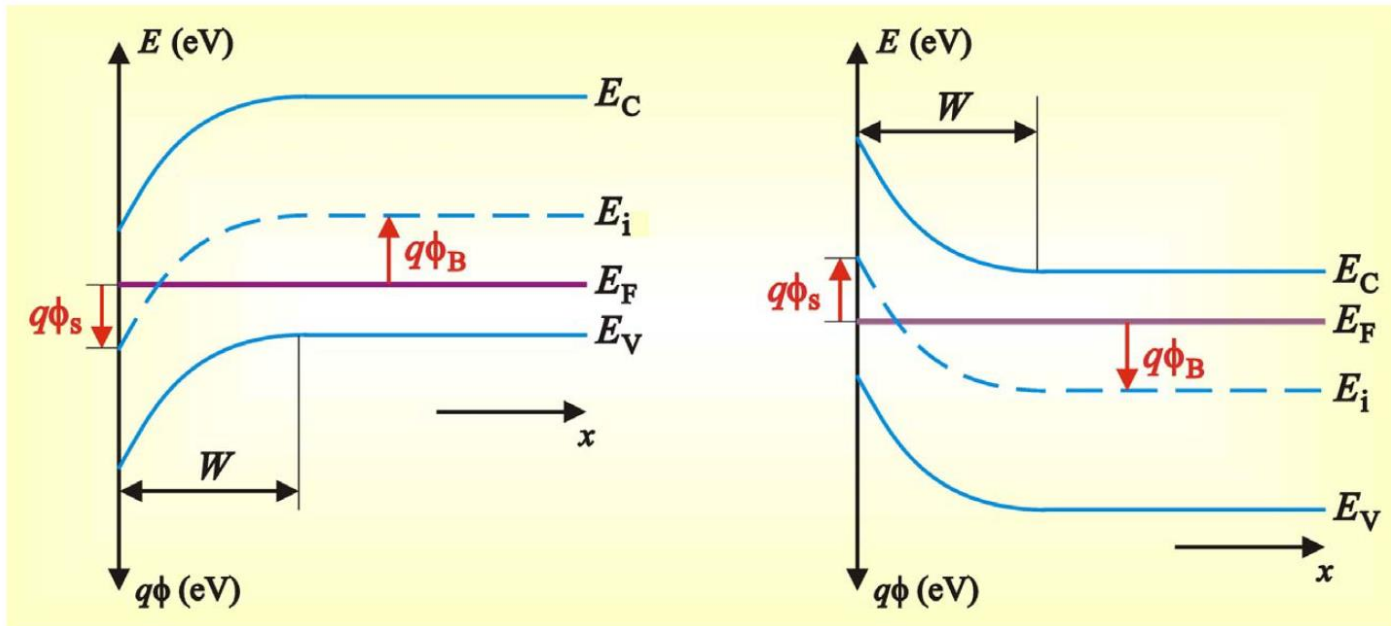
$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = -\frac{\rho(x, y, z)}{\epsilon}$$

- $\phi$  - potential
- $\rho$  - net charge density per unit volume
- $\epsilon$  - permittivity of the semiconductor medium

**A set of simplifying assumptions would be needed to obtain an analytical expression for the potential.**

# Coordinate System

$x = 0$  is at the Si-SiO<sub>2</sub> interface



Left : Energy band diagram of a p-type Si substrate

Right : Energy band diagram of an n-type Si substrate

# Definitions of Potentials

- Reference: Fermi level  $E_F$
- Potential at point  $x$ :

$$\phi(x) = \frac{E_F - E_i(x)}{q}$$

- Semiconductor bulk potential:

$$\phi_B = \frac{E_F - E_i(x \rightarrow \infty)}{q}$$



# Semiconductor Bulk Potential

From:

- P-type Si substrate:

$$p = n_i \exp[(E_i - E_F) / kT] \approx N_A$$

$$\phi_B = -\frac{kT}{q} \ln\left(\frac{N_A}{n_i}\right)$$

From:

- N-type Si substrate:

$$n = n_i \exp[(E_F - E_i) / kT] \approx N_D$$

$$\phi_B = \frac{kT}{q} \ln\left(\frac{N_D}{n_i}\right)$$

# Assumptions

- One-dimensional approach
  - Solve Poisson's equation in only one direction, along the depth of the Si substrate ( $x$ -direction).
  - Ignore fringing fields at the edge of the MOS capacitor (valid for large geometry where the electric field is largely perpendicular to the Si-SiO<sub>2</sub> interface).

$$\Rightarrow \frac{\partial^2 \phi}{\partial x^2} = -\frac{\rho(x)}{\epsilon}$$

# Assumptions

- Depletion approximation
  - The concentrations of mobile charges in the depletion region are assumed to be much smaller compared to that of the ionized impurity atoms.

$$\begin{aligned}\rho(x) &= q[p(x) + N_D(x) - n(x) - N_A(x)] \\ &\approx q[N_D(x) - N_A(x)]\end{aligned}$$

$$\therefore \frac{\partial^2 \phi}{\partial x^2} = \frac{q[N_A(x) - N_D(x)]}{\epsilon}$$

# Assumptions

- Uniformly doped Si substrate
  - To further simplify the integral, the doping concentration is assumed to be constant along the depth of the Si substrate.
  - In addition, we define a net doping concentration given by  $N_{\text{B}} = N_{\text{A}} - N_{\text{D}}$ .

$$\therefore \frac{\partial^2 \phi}{\partial x^2} = \frac{qN_{\text{B}}}{\epsilon}$$

# Approximate Solution

- Electric field:

Recall:

$$\xi(x) = -\partial\phi/\partial x$$

$$\xi(x) = -\frac{\partial\phi}{\partial x} = -\left(\frac{qN_B}{\epsilon} x + C_0\right)$$

- Potential:

$C_0$  and  $C_1$  are constants of integration.

$$\phi(x) = \frac{qN_B}{2\epsilon} x^2 + C_0 x + C_1$$

# Approximate Solution

- **Boundary conditions:**

- At  $x = 0$  (Si-SiO<sub>2</sub> interface):

$$\phi(x = 0) = \phi_s$$

- At  $x = W$  (depletion region edge):

$$\phi(x = W) = \phi_B$$

$$\xi(x = W) = 0$$

# Approximate Solution

From:

- Constant  $C_0$ :

$$\xi(x=W) = -\left(\frac{qN_B W}{\epsilon} + C_0\right) = 0$$

$$C_0 = -\frac{qN_B W}{\epsilon}$$

From:

- Constant  $C_1$ :

$$\phi(x=W) = \frac{qN_B W^2}{2\epsilon} - \frac{qN_B W^2}{\epsilon} + C_1 = \phi_B$$

$$C_1 = \phi_B + \frac{qN_B W^2}{2\epsilon}$$

# Approximate Solution

- **Potential:**

The potential is a quadratic function of distance.

$$\begin{aligned}\phi(x) &= \frac{qN_B}{2\epsilon} (x^2 - 2Wx + W^2) + \phi_B \\ &= \frac{qN_B}{2\epsilon} (x - W)^2 + \phi_B\end{aligned}$$

- **Band bending:**

$$\psi_s = \phi_s - \phi_B = \frac{qN_B W^2}{2\epsilon}$$



# Approximate Solution

- Electric field:

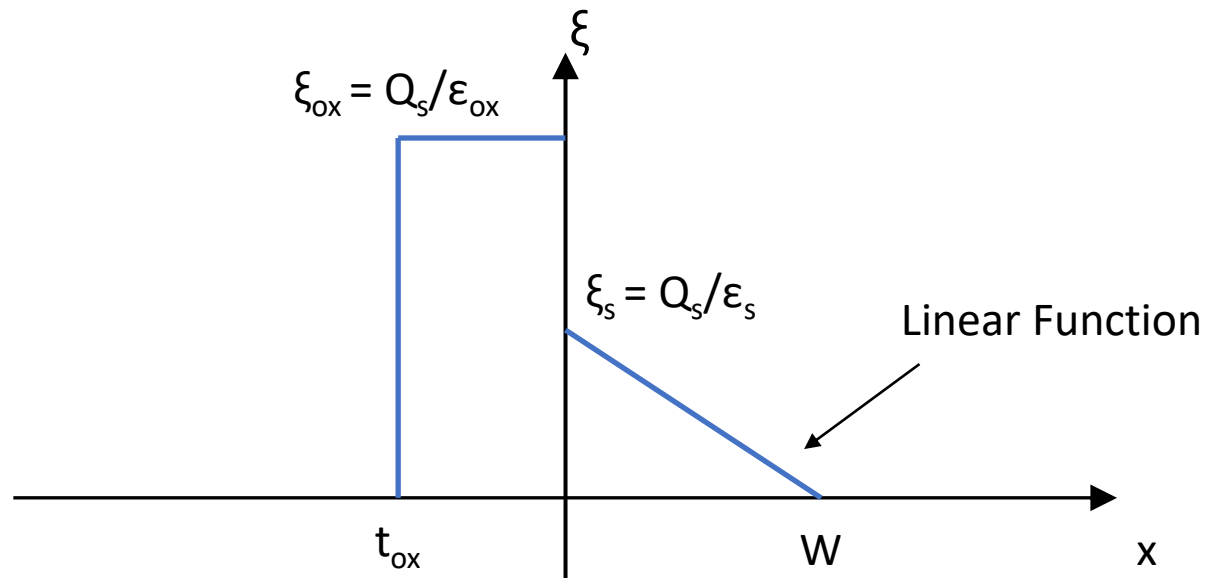
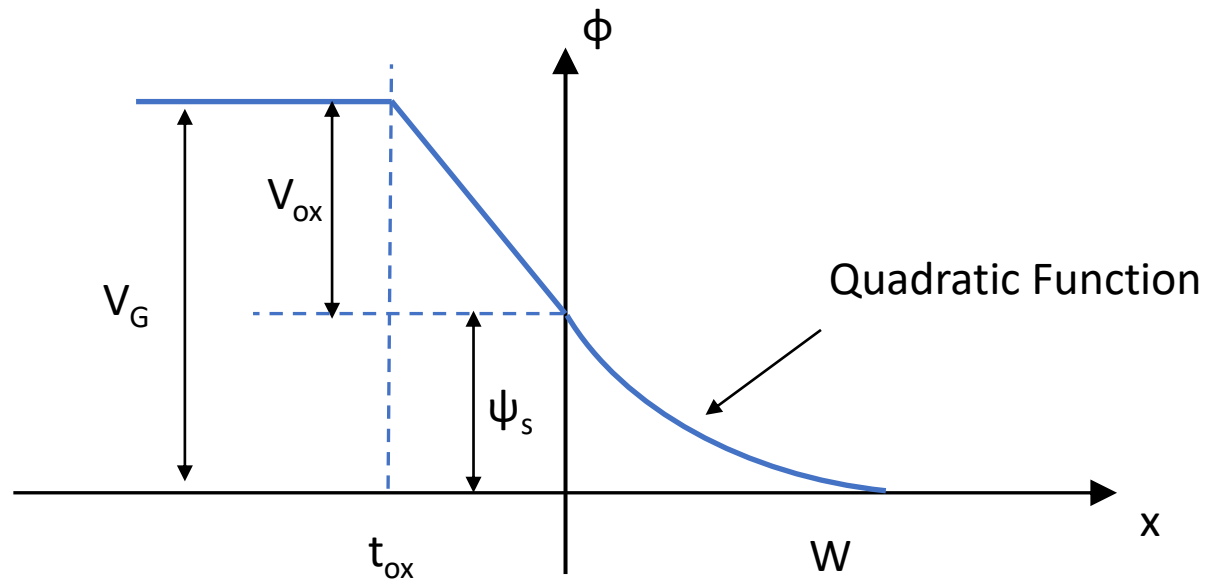
The electric field is a linear function of distance.

$$\xi(x) = -\frac{\partial\phi}{\partial x} = \frac{qN_B}{\epsilon}(W - x)$$

- Depletion width:

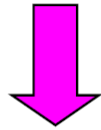
The depletion width is directly proportional to  $(\psi_s)^{1/2}$ .

$$W = \sqrt{\frac{2\epsilon\psi_s}{qN_B}}$$



# $V_G - \psi_s$ Relation

Voltage drop across  
the oxide layer



$$V_G = V_{ox} + \psi_s$$



Voltage drop across  
the Si substrate

# $V_G - \psi_s$ Relation

- Voltage drop across the oxide:

$$V_{ox} = - \int_0^{-t_{ox}} \xi_{ox}(x) dx = \xi_{ox} t_{ox}$$

- $\xi_{ox}$  is the electric field in the oxide. Note that  $\xi_{ox}$  is a constant for a charge-free oxide layer.
- $t_{ox}$  is the thickness of the oxide layer.

**Objective:** To relate  $\xi_{ox}$  to the Si surface field  $\xi_s$  and band bending  $\psi_s$ .

# $V_G - \psi_s$ Relation

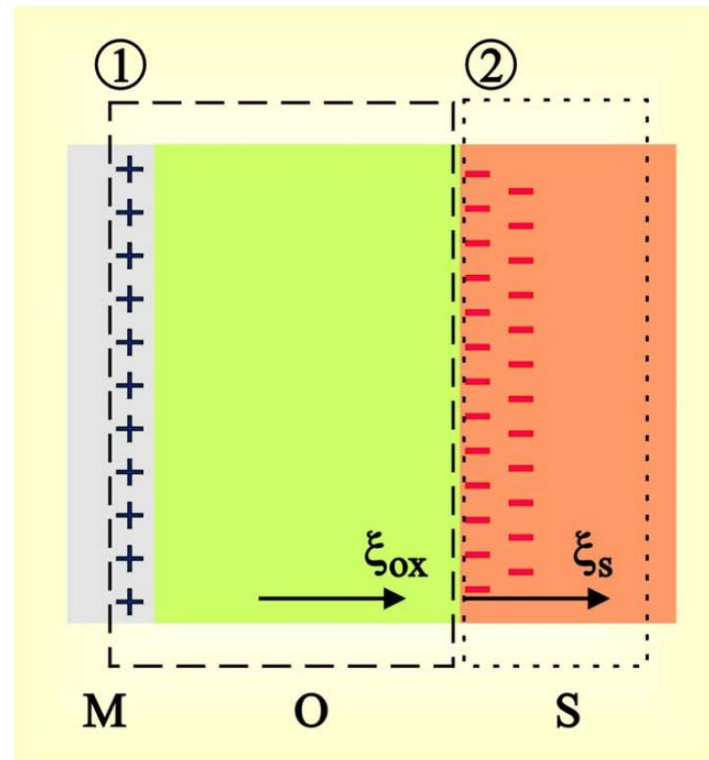
- Gauss's law:

$$\oint \xi \cdot \partial S = \frac{Q}{\epsilon}$$

- $\partial S$  is a vector perpendicular to the Gaussian surface.
- The Gaussian surface refers to any arbitrarily-shaped hypothetical surface that encloses the charge of interest.
- The circle on the integral sign indicates a closed-surface integration.

# $V_G - \psi_s$ Relation

- Gaussian surfaces:
  - Surface 1 extends from  $x = 0^-$  to  $x = -t_{ox}^+$ .
  - Surface 2 extends from  $x = 0^+$  to  $+\infty$ .



## $V_G - \psi_s$ Relation

- Gaussian surface 1:

$$\epsilon_{ox} \xi_{ox} = Q_g$$

- Gaussian surface 2:

$$-\epsilon_s \xi_s = Q_s$$

- The negative sign accounts for the fact that the Si surface electric field  $\xi_s$  is pointing in a direction opposite to the normal surface vector.

## $V_G - \psi_s$ Relation

- From charge neutrality consideration,

$$\epsilon_{ox} \xi_{ox} = \epsilon_s \xi_s$$

$$\Rightarrow \xi_{ox} = \frac{\epsilon_s \xi_s}{\epsilon_{ox}}$$

$$\therefore V_{ox} = \xi_{ox} t_{ox} = \frac{\epsilon_s \xi_s}{\epsilon_{ox}} t_{ox} = -\frac{Q_s}{C_{ox}}$$



## $V_G - \psi_s$ Relation

- From the depletion approximation approach, the charge density per unit area in the Si substrate in depletion is given as  $Q_s = -qN_B W$ .
- The corresponding  $V_G - \psi_s$  relation is

$$V_G = \psi_s \pm \frac{\sqrt{2q\epsilon_s N_B \psi_s}}{C_{ox}}$$

- Negative sign for n-type substrate; positive sign for p-type substrate.

# Threshold Voltage

- Physical definition:

$$\phi_s = -\phi_B ; \quad \psi_s = -2\phi_B$$

- Silicon charge at onset of inversion:

$$Q_s = Q_i + Q_B$$

- $Q_i$  is the inversion charge density per unit area.
- $Q_B$  is the bulk charge density per unit area.

# Threshold Voltage

- Since the inversion layer thickness is much smaller than the depletion region,

$$Q_i \ll Q_B; \quad Q_s \approx Q_B$$

- Applying the criterion  $\psi_s = -2\phi_B$ , an expression for threshold voltage results:

$$V_{TH} = -2\phi_B \pm \frac{\sqrt{-4\epsilon q N_B \phi_B}}{C_{ox}}$$

# MOS Capacitance

- **Definition:**

$$C \equiv \frac{\partial Q_G}{\partial V_G} \equiv -\frac{\partial Q_s}{\partial V_G}$$

- $Q_G$  - charge per unit area on the gate
- $Q_S$  - charge per unit area in the Si substrate

**Note:**

$Q_G + Q_S = 0$  or  $Q_G = -Q_S$  is always valid, by the charge conservation principle.

# MOS Capacitance

- Equivalent circuit representation:

$$\frac{1}{C} = \frac{\partial V_G}{\partial Q_G} = \frac{\partial V_{ox}}{\partial Q_G} + \frac{\partial \psi_s}{-\partial Q_s}$$

Note:

$$\delta V_G = \delta V_{ox} + \delta \psi_s$$

- Gate oxide capacitance:  $C_{ox} = \partial Q_G / \partial V_{ox}$
- Silicon capacitance :  $C_s = -\partial Q_s / \partial \psi_s$

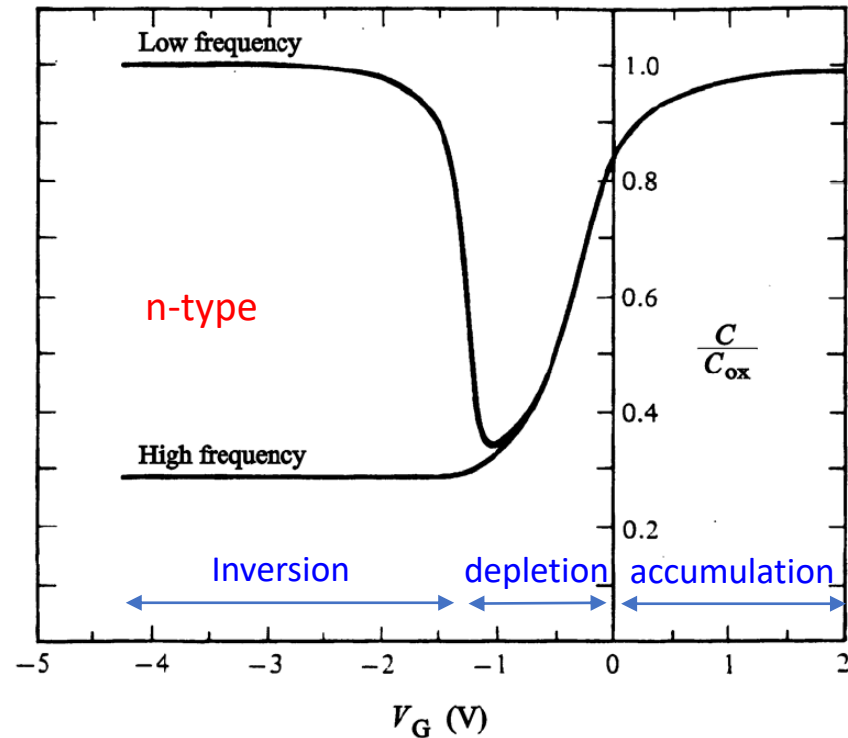
$$\frac{1}{C} = \frac{1}{C_{ox}} + \frac{1}{C_s}$$

# MOS Capacitance – Carrier Response

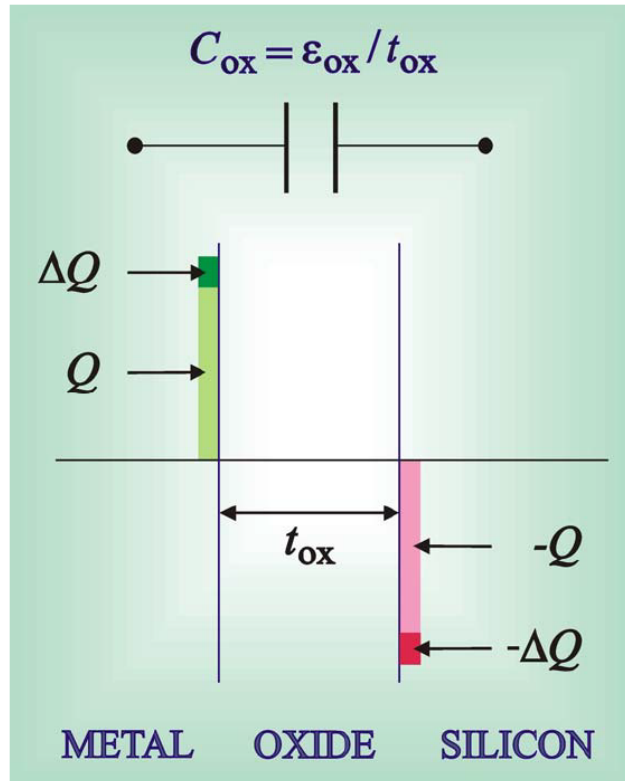
- Majority carriers:
  - Readily supplied / removed through the back contact.
  - Response time is in the range of  $1\mu\text{s}$  –  $0.1\text{ps}$ .
- Minority carriers:
  - Supplied by thermal generation; removed through recombination with majority carriers.
  - Response time is inversely proportional to the bulk trap density of the Si substrate; can be as slow as several hundreds of milliseconds for state-of-the-art device grade Si.

# Capacitance-Voltage Curves

Typical high and low frequency C-V characteristics of an MOS capacitor. The high frequency curve was derived from a measurement frequency of 1 MHz. The low frequency curve was obtained employing the quasi-static technique. The substrate doping concentration  $N_D = 9.1 \times 10^{14} \text{ cm}^{-3}$ , and oxide thickness is  $0.119 \mu\text{m}$ .



# MOS Capacitance - Accumulation



(n-type)

- Majority carriers in the accumulation layer respond to the ac signal.
- Silicon capacitance:

$$Q_s \propto \exp\left(-\frac{q\psi_s}{2kT}\right)$$

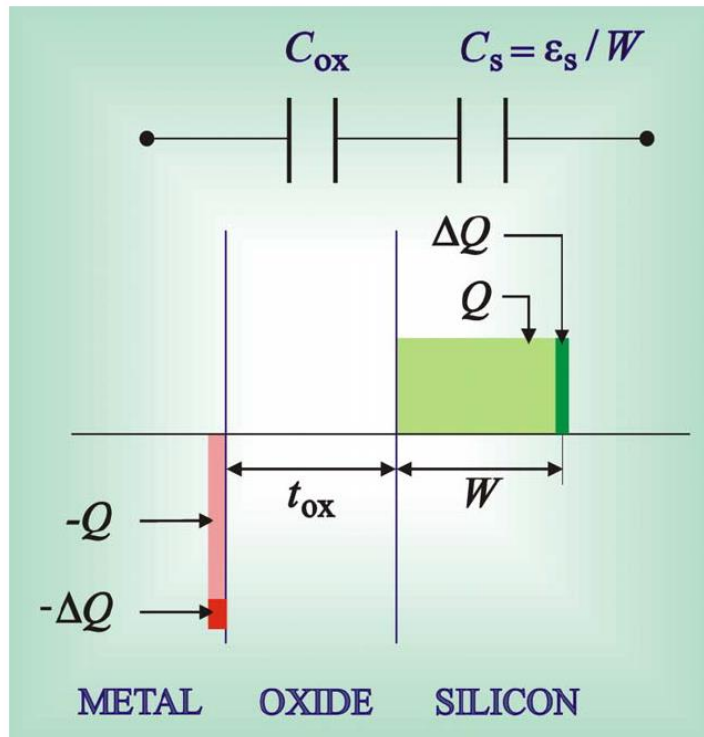
$$\Rightarrow C_s = -\frac{\partial Q_s}{\partial \psi_s} = \frac{qQ_s}{2kT}$$

- “Parallel plate” capacitance

$$\because C_s \gg C_{ox} \Rightarrow C \approx C_{ox}$$



# MOS Capacitance - Depletion

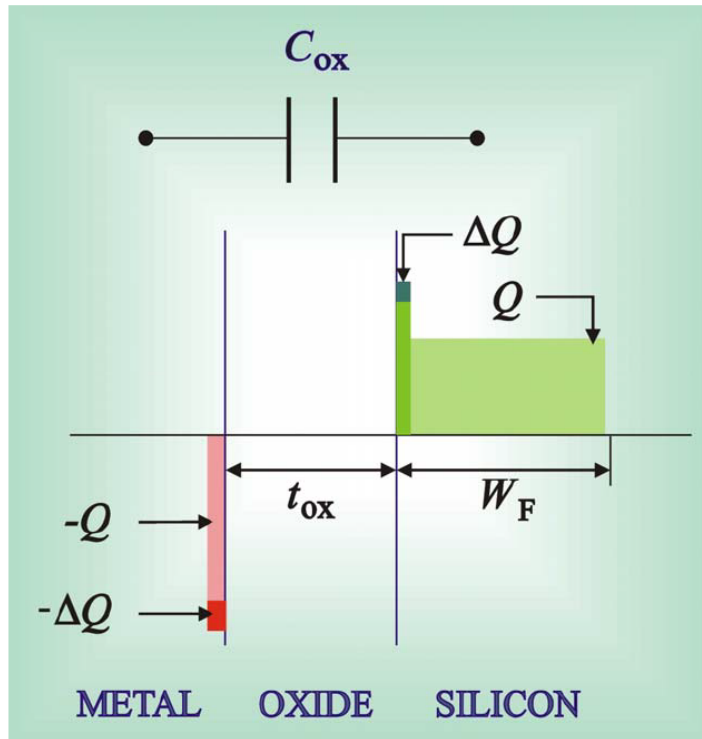


- Majority carriers at the depletion edge respond to the ac signal.
- The “plates” are separated by two insulating layers.
- Silicon capacitance:

$$C_s = \frac{\epsilon_s}{W}$$

$$\frac{1}{C} = \frac{1}{C_{ox}} + \frac{1}{C_s}$$

# MOS Capacitance - Inversion

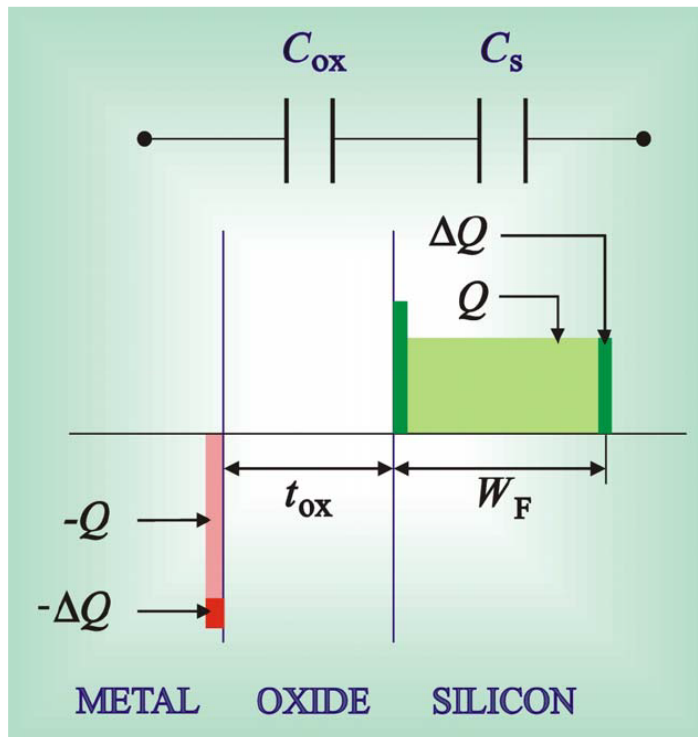


- If the frequency of the ac signal is low, then minority carriers could be thermally generated or recombined in time to meet the charging or discharging of the inversion layer.

# MOS Capacitance - Inversion

- In strong inversion, the surface mobile charges completely screen the rest of the Si substrate from the influence of the ac voltage.
- The result is a “parallel-plate” gate oxide capacitance, with one plate at the inversion layer, and the other at the gate.

# MOS Capacitance - Inversion



- If a high-frequency ac signal is used, the minority carriers in the inversion layer would not be able to follow the change in the ac voltage.
- Majority carriers at the edge of the depletion region would respond to maintain overall charge neutrality.

# MOS Capacitance - Inversion

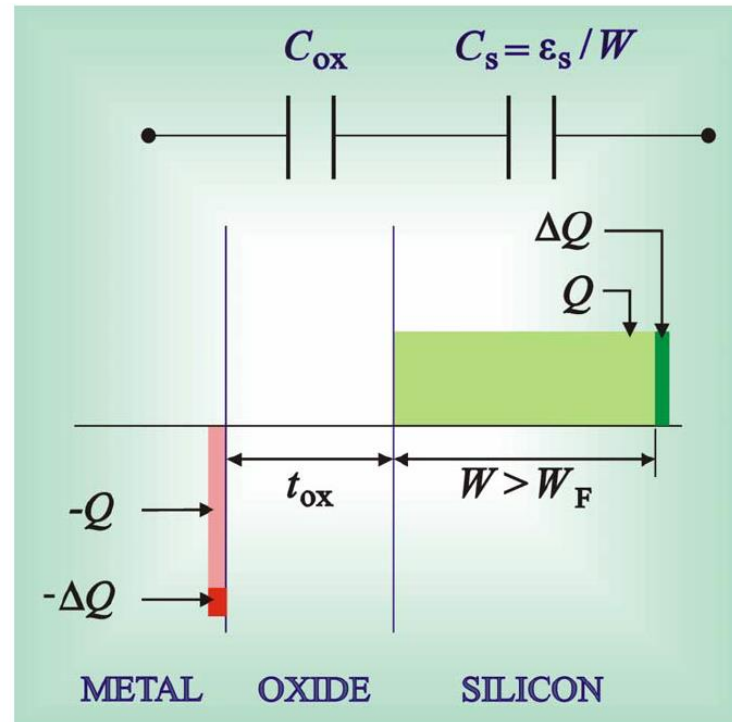
- The overall capacitance is given by the series combination of  $C_{ox}$  and  $C_s (= \epsilon_s/W)$ .
- One interesting point:
  - Since the depletion width does not increase much with a further increase in the gate voltage beyond inversion, the Si depletion capacitance  $C_s$  is therefore relatively independent of the gate bias.
  - This explains the “plateau” observed in the figure. The resultant C-V curve is commonly known as the **high frequency C-V** curve.

# MOS Capacitance - Deep Depletion

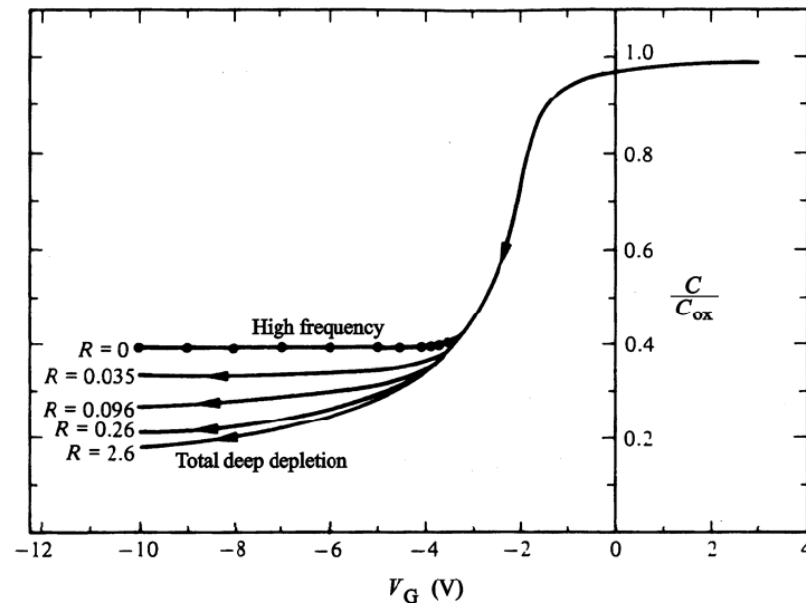
- In previous discussion, we have assumed that the dc gate bias is changed sufficiently slowly that the Si substrate remains in thermal equilibrium.
- If the gate sweep rate is so fast that thermal generation of minority carriers is unable to keep up with the demand, then majority carriers at the depletion edge would be repelled to uncover more ionized impurity atoms in order to achieve charge neutrality.

# MOS Capacitance - Deep Depletion

- When this happens, the MOS capacitor is said to have entered deep depletion.
- The deep depletion mode is characterized by a depletion width larger than  $W_F$ , and a capacitance lower than  $C_F$ .



# MOS Capacitance - Deep Depletion

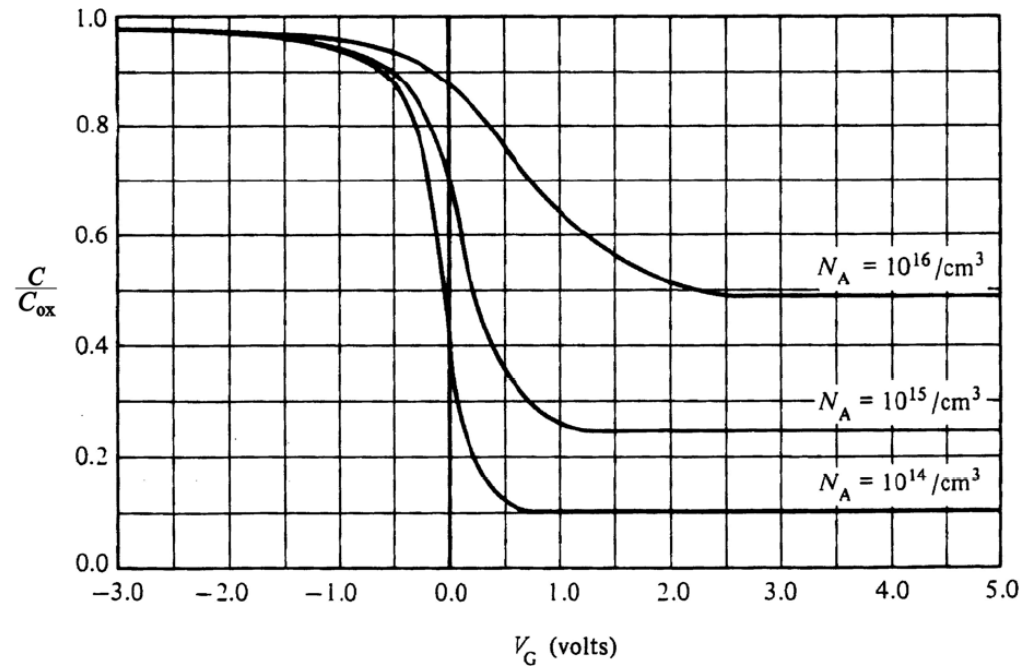


Measured C-V characteristics as a function of the ramp rate ( $R$ ). In inversion, the high frequency capacitance was obtained by stopping the ramp and allowing the device to equilibrate.



# C-V Curve

p-type



High frequency C-V characteristics for MOS capacitors with different substrate doping concentrations.

$$\frac{1}{C} = \frac{1}{C_{ox}} + \frac{1}{C_s}$$

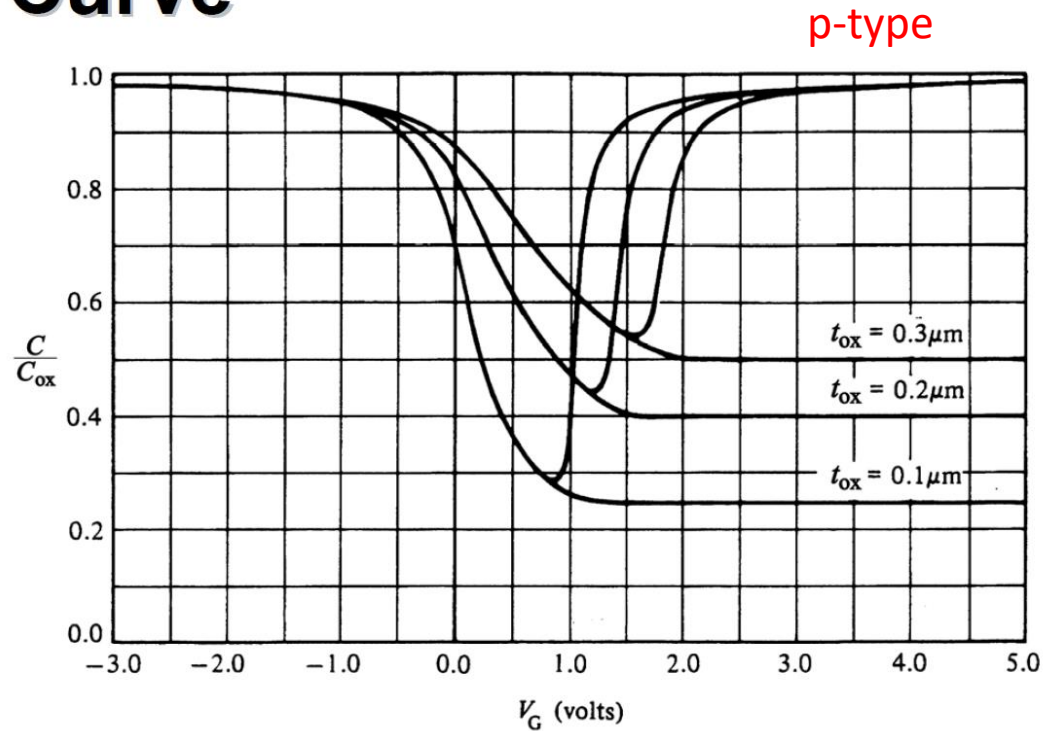
$$C = \frac{C_{ox}C_s}{C_{ox} + C_s}, \quad \frac{C}{C_{ox}} = \frac{C_s}{C_{ox} + C_s}, \quad \frac{C}{C_{ox}} = \frac{1}{\frac{C_{ox}}{C_s} + 1}$$

Assume that  $t_{ox}$  is fixed, and so  $C_{ox}$  is constant.

$$C_s = \frac{\epsilon_s}{W} \text{ and } W = \sqrt{\frac{2\epsilon_s\psi_s}{qN_A}}$$

When  $N_A$  increases,  $W$  decreases,  $C_s$  increases,  $C_{ox}/C_s$  decreases,  $C/C_{ox}$  increases.

# C-V Curve

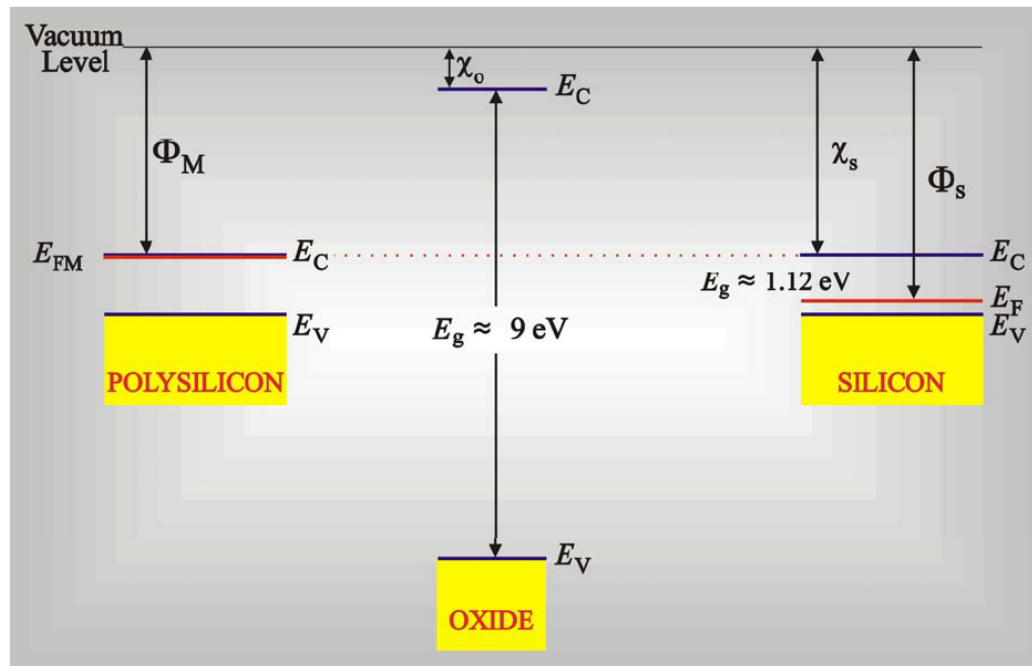


Low and high frequency C-V characteristics of MOS capacitors with different gate oxide thicknesses.

# C-V Curve

- Questions:
  - Explain the dependence of  $C/C_{ox}$  on substrate doping concentration and oxide thickness, i.e. why does  $C/C_{ox}$  increase with substrate doping concentration and oxide thickness in the depletion and inversion regions?

# Non-ideality: Work function Difference



Individual energy band diagrams of the metal ( $n^+$  polysilicon),  $\text{SiO}_2$ , and p-Si substrate of the MOS structure. The work function of the  $n^+$  polysilicon is less than that of the p-Si.

## Non-ideality: Work function Difference

- What happens when  $\Phi_M \neq \Phi_s$ ?
  - Consider a typical example of an  $n^+$  polysilicon gate and a p-type substrate.
  - The Fermi level of the  $n^+$  polysilicon gate is close to  $E_C$ . In our analysis, we assume  $E_F = E_C$ .
  - Hence, the work function of the gate,  $\Phi_M = \chi_s$ , while that of the p-Si substrate is  $\chi_s + (E_C - E_F) > \Phi_M$ .
  - What is the implication when  $E_{Fm} > E_F$ ?

## Non-ideality: Work function Difference

- Conceptually, we may interpret the condition  $E_{\text{FM}} > E_{\text{F}}$  as:
  - Electrons in the  $\text{n}^+$  polysilicon gate have an average energy higher than those in the p-Si substrate.
- Thus, when the three components of the MOS system are brought into contact, a net transfer of electrons from the  $\text{n}^+$  gate to the p-Si occurs.
  - This transfer continues until the average energy of electrons is the same throughout the MOS system, i.e. when the  $E_{\text{FM}}$  is aligned with  $E_{\text{F}}$ .

## Non-ideality: Work function Difference

- The departure of electrons results in a net **positive** charge on the gate, while the arrival of electrons leads to a net **negative** charge at the surface of the p-Si substrate.

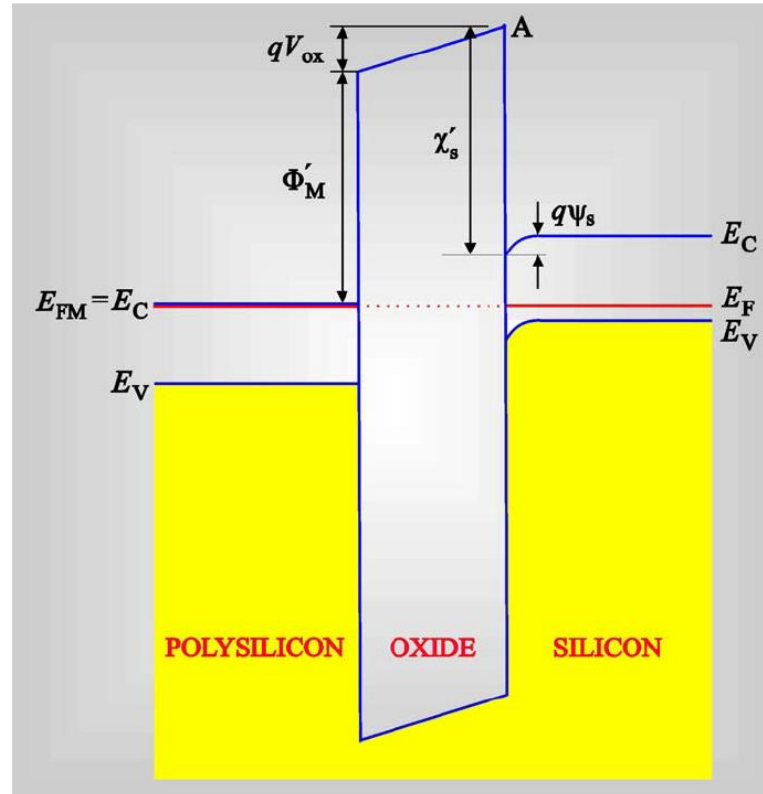
**A built-in electric field or potential thus exists in the MOS capacitor in thermal equilibrium!**



# Non-ideality: Work function Difference

Energy band diagram of the  $n^+$  polysilicon/ $\text{SiO}_2$ /p-Si capacitor in thermal equilibrium.

This configuration results in a net negative charge at the Si surface.



# Non-ideality: Work function Difference

- Recall the  $V_G - \psi_s$  relation:

$$V_G = V_{ox} + \psi_s$$

- Clearly, this relation is no longer valid in this case, since  $V_{ox} + \psi_s > 0$  even when  $V_G = 0$ .
- We need to modify the expression by adding a suitable term on the LHS of the equation:

$$V_G - \phi_{MS} = V_{ox} + \psi_s$$

Why  $-\phi_{MS}$ ?

$\phi_{MS} = (\Phi_M' - \Phi_s')/q$  is the work function difference

# Non-ideality: Work function Difference

- Energy at point A with reference to  $E_{\text{FM}}$ :

$$E_{\text{A}} = \Phi_{\text{M}}' + qV_{\text{ox}}$$

- Energy at point A with reference to  $E_{\text{F}}$ :

$$E_{\text{A}} = (E_{\text{C}} - E_{\text{F}})_{x \rightarrow \infty} - q\psi_{\text{s}} + \chi_{\text{s}}'$$

- Equating:

$$\Phi_{\text{M}}' + qV_{\text{ox}} = (E_{\text{C}} - E_{\text{F}})_{x \rightarrow \infty} - q\psi_{\text{s}} + \chi_{\text{s}}'$$

$$q(V_{\text{ox}} + \psi_{\text{s}}) = (E_{\text{C}} - E_{\text{F}})_{x \rightarrow \infty} + \chi_{\text{s}}' - \Phi_{\text{M}}' = -\phi_{\text{MS}}$$

## Non-ideality: Work function Difference

- Note: For a p-type substrate,  $\phi_{MS}$  is **negative** for an  $n^+$  polysilicon gate, but **positive** for a  $p^+$  polysilicon gate.
- Question:
  - Draw the energy band diagram, at thermal equilibrium, of an  $n^+$  polysilicon/ $\text{SiO}_2$ /n-Si capacitor. What is the corresponding charge state of the Si surface?

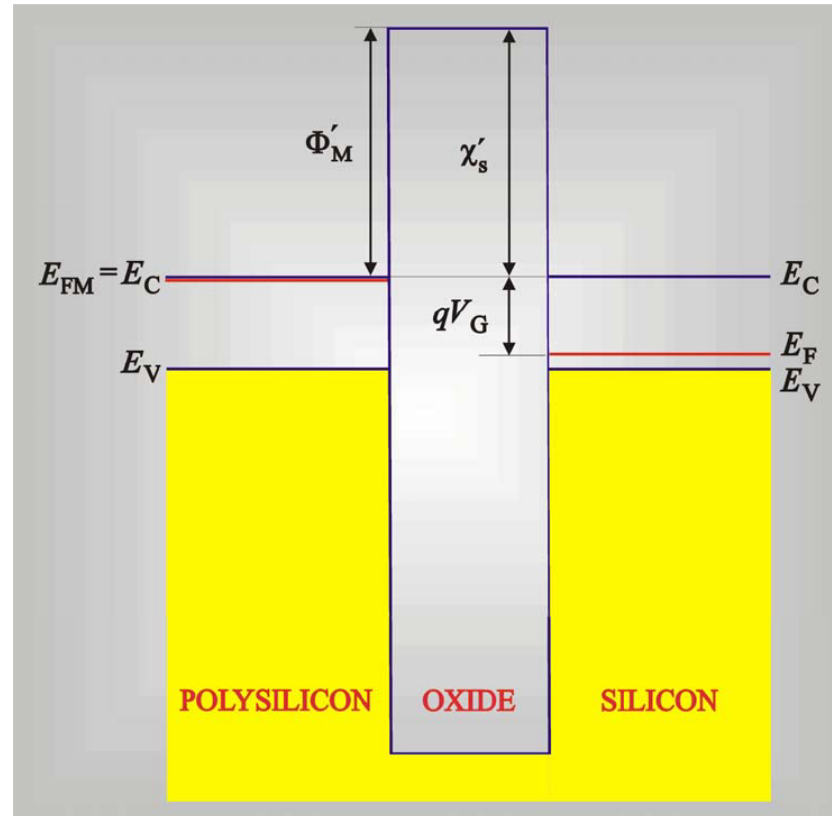
# Concept of Flat band Voltage

- Flat band voltage:
  - The gate voltage needed to achieve a neutral (zero band bending) condition throughout the Si substrate, i.e.  $\psi_s = 0$ .
  - In the case where a non-zero work function exists,

$$V_{\text{FB}} = \phi_{\text{MS}}$$

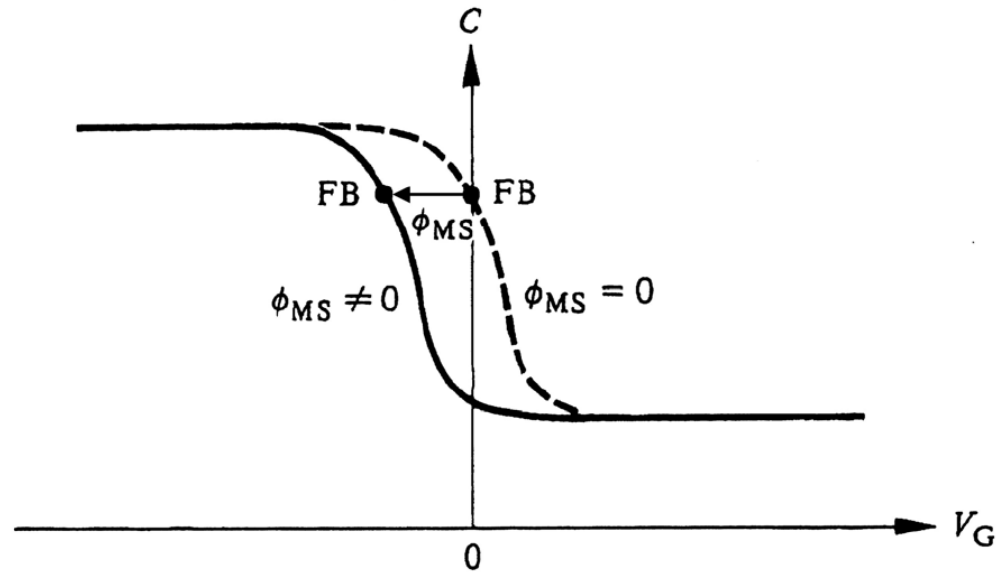
# Concept of Flat band Voltage

Schematic diagram illustrating the application of a gate voltage equal to the work function difference, to achieve flat band condition in the Si substrate. The corresponding gate voltage is also known as the flat band voltage.



# Non-ideality: Work function Difference

- Effect on C-V curve:



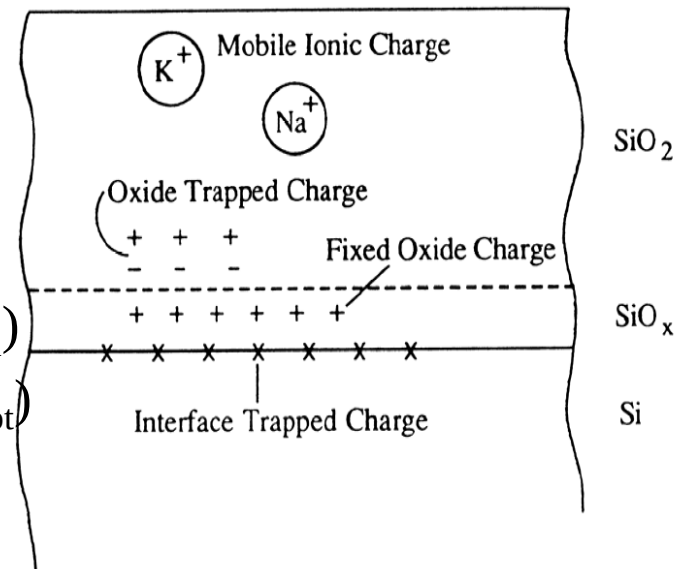
# Non-ideality: Oxide/Interface Trapped Charge

- Defects exist in practical  $\text{SiO}_2$  layers.
  - **Structural defects**
    - These result in energy levels within the  $\text{SiO}_2$  band gap. These energy levels can provide permanent resting sites for electrons/holes.
  - **Trapped Charge**
    - The confinement of electrons/holes within the  $\text{SiO}_2$  layer. These charges are said to be “trapped” at energy levels relating to oxide structural defects, which are either introduced during fabrication or device operation.



## Non-ideality: Oxide/Interface Trapped Charge

- In general, there are **four** different types of charges in the  $\text{SiO}_2$ :
  - Fixed oxide charge ( $Q_f$ )
  - Mobile oxide charge ( $Q_m$ )
  - Oxide trapped charge ( $Q_{ot}$ )
  - Interface traps ( $Q_{it}$ )



Charges and their location in thermally oxidized  $\text{SiO}_2$ .

The result is to increase the overall negative value of  $V_{FB}$ .

$$V_{FB} = \phi_{ms} - \frac{(Q_f + Q_m + Q_{ot})}{C_{ox}}$$

The effect is to shift the C-V curve to the left by an amount equal to the magnitude of  $V_{FB}$ .

# Non-ideality – Polysilicon Depletion

- Implicit assumption:
  - The polysilicon gate is degenerately doped so that it basically function like a perfect conductor, with no voltage drop across it.
  - This is valid for doping concentration on the order of  $10^{20} \text{ cm}^{-3}$  and higher.
- Modern technology:
  - Insufficient dopant activation due to low temperature processing.
  - Outdiffusion of dopants into silicide layer.

# Non-ideality – Polysilicon Depletion

- Consequence:
  - The polysilicon gate depletes/inverts just like the Si substrate.
  - This depletion/inversion is necessary to produce sufficient charge of opposite polarity to compensate the charge in the Si substrate.
    - Consider a p-type Si substrate. In inversion, the charge in the Si is negative. This is balanced by an equal charge of opposite polarity on the gate. If the polysilicon gate is doped n-type, it needs to deplete/invert to produce the required positive charge (holes + ionized donor atoms).

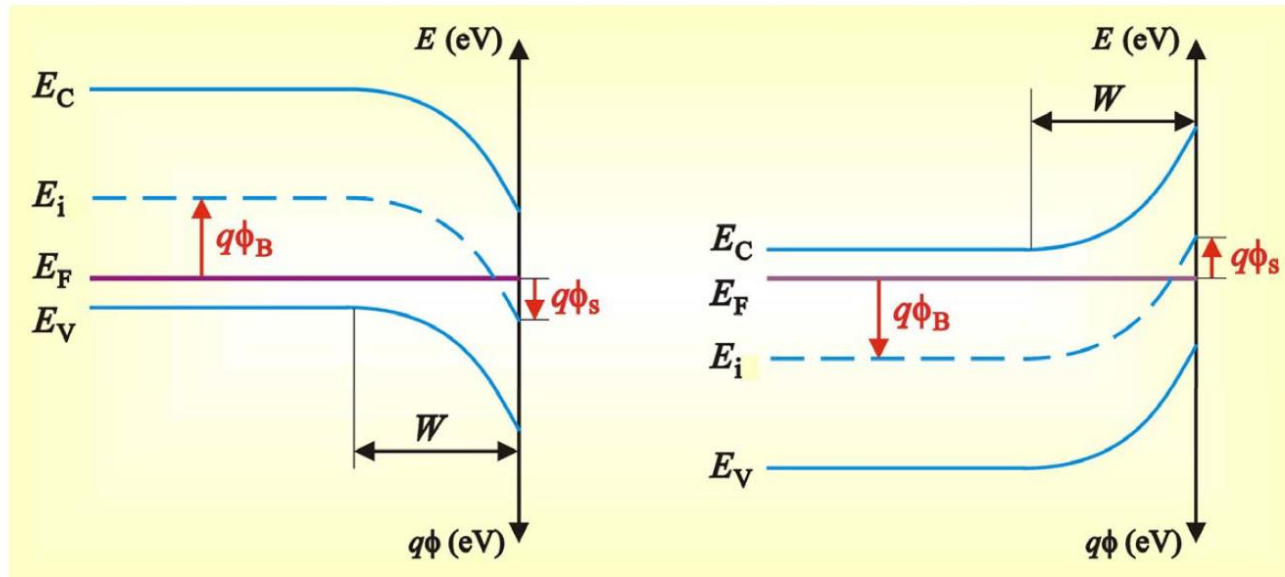
# Non-ideality – Polysilicon Depletion

- Consequence (cont'd):
  - Polysilicon depletion/inversion implies a voltage drop across the gate.
  - This additional voltage drop has to be accounted for when calculating the threshold voltage of the device.

$$V_G = V_{FB} + \psi_s - \frac{Q_s}{C_{ox}} + V_{poly} (= -\psi_{poly})$$

- $V_{poly}$  can be calculated by considering charge neutrality and applying the electrostatics already established for the Si substrate.

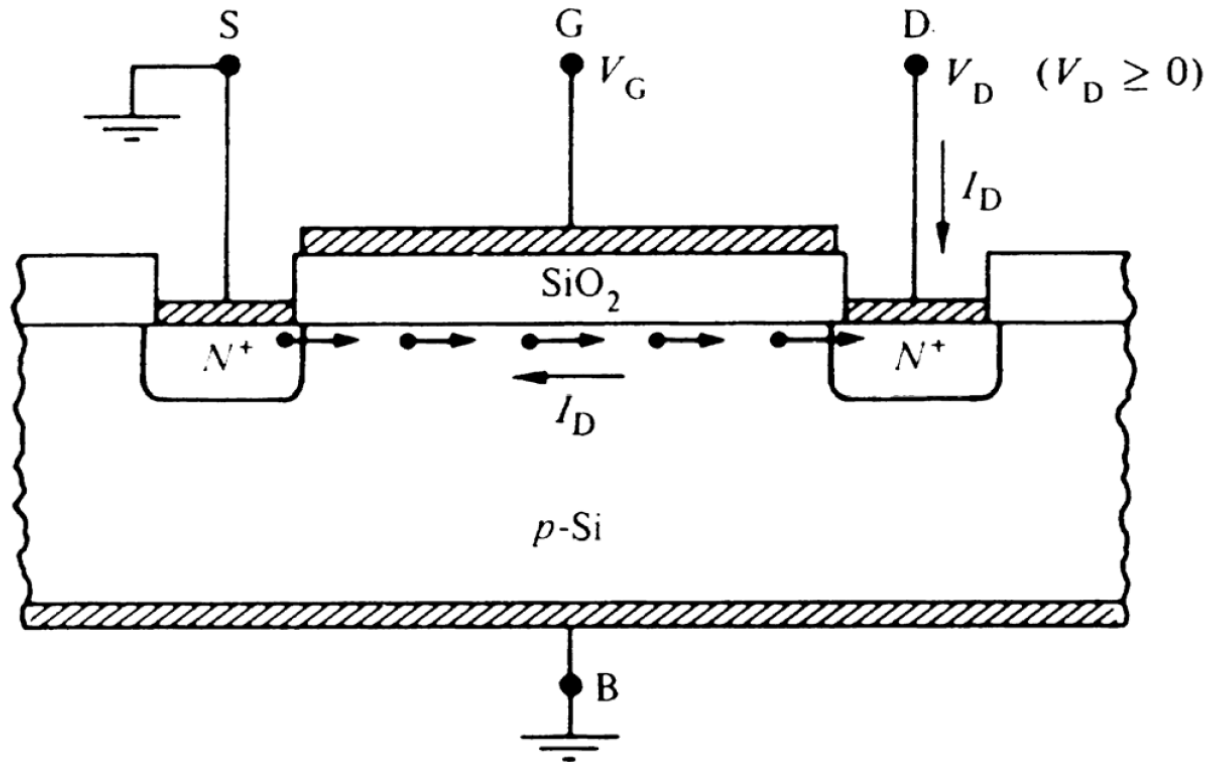
# Non-ideality – Polysilicon Depletion



$$V_{\text{poly}} = \phi_B - \phi_s = -\psi_{\text{poly}}$$

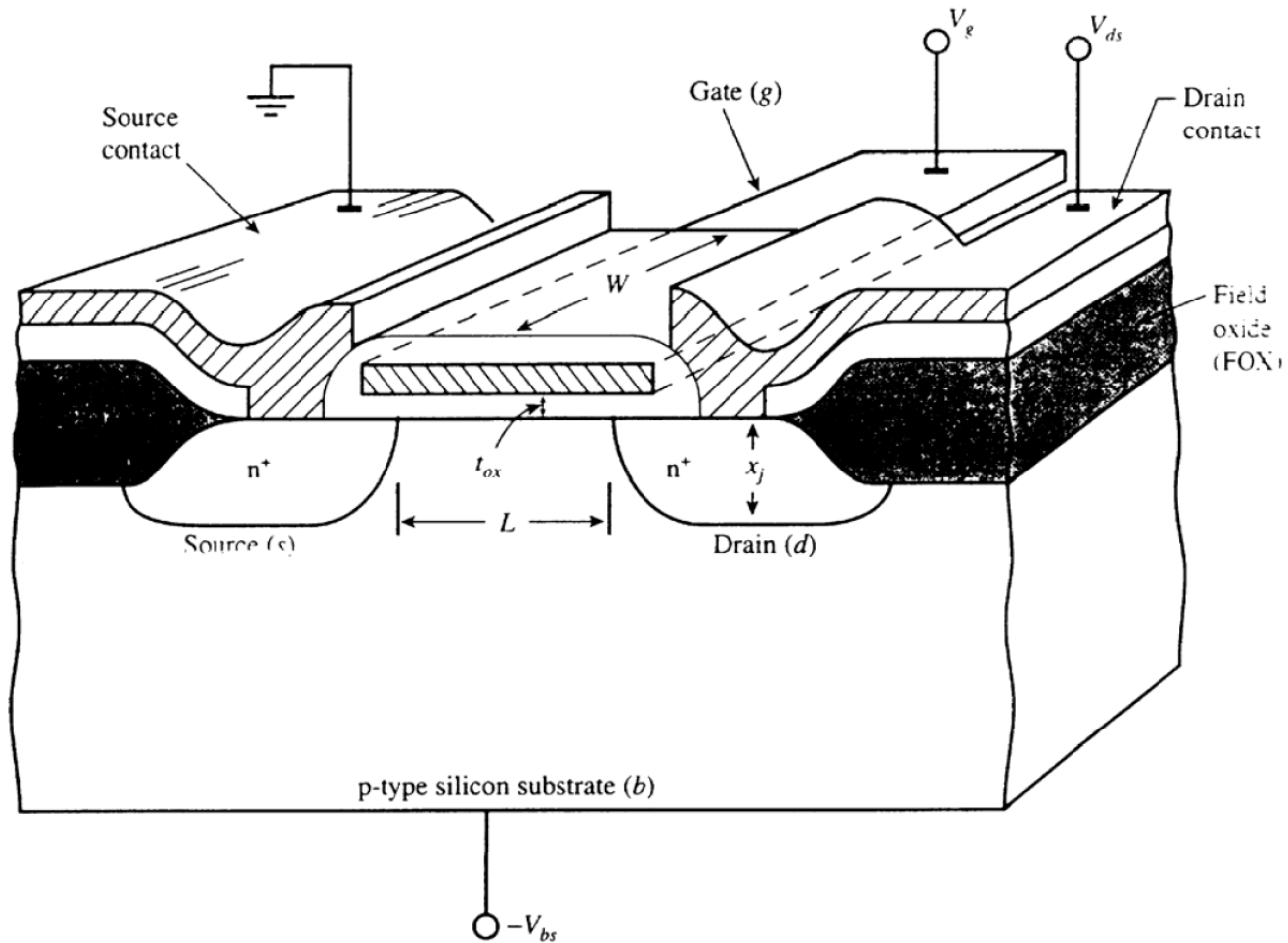
# MOSFET Fundamentals

# MOSFET Structure



Cross-sectional view of the basic  $p$ -bulk (n-channel) MOSFET structure, showing the terminal designations, carrier and current directions, and standard biasing conditions.





Three-dimensional view of the basic MOSFET device structure.

- ❑ Physically, the MOSFET is an MOS capacitor with two p-n junctions placed immediately adjacent to the region of the semiconductor controlled by the gate (G) electrode (normally n<sup>+</sup> polysilicon).
- ❑ The semiconductor substrate is either p-type or n-type
  - For a p-type substrate, the two islands are n<sup>+</sup> doped.
  - For an n-type substrate, the two islands are p<sup>+</sup> doped.
- ❑ One n<sup>+</sup> region is known as the source electrode (S), where carriers are injected, and the other is known as the drain electrode (D), where carriers (electrons) are collected or extracted.

- ❑ Charge carriers enter the MOSFET through the source and leaves through the drain, and their flow is subjected to the control or gating action of the gate.
- ❑ The drain current  $I_D$  is taken to be positive when flowing into the drain electrode.
- ❑ Standard biasing condition: The source and substrate are connected to the common [ $V_S = V_{BS} = 0$ ], the drain-substrate junction is reverse-biased ( $V_{DS} > 0$ ).
- ❑ For n-MOSFET (p-MOSFET), a positive (negative) voltage  $V_{GS}$  is applied to the gate.

# Types of MOSFETs

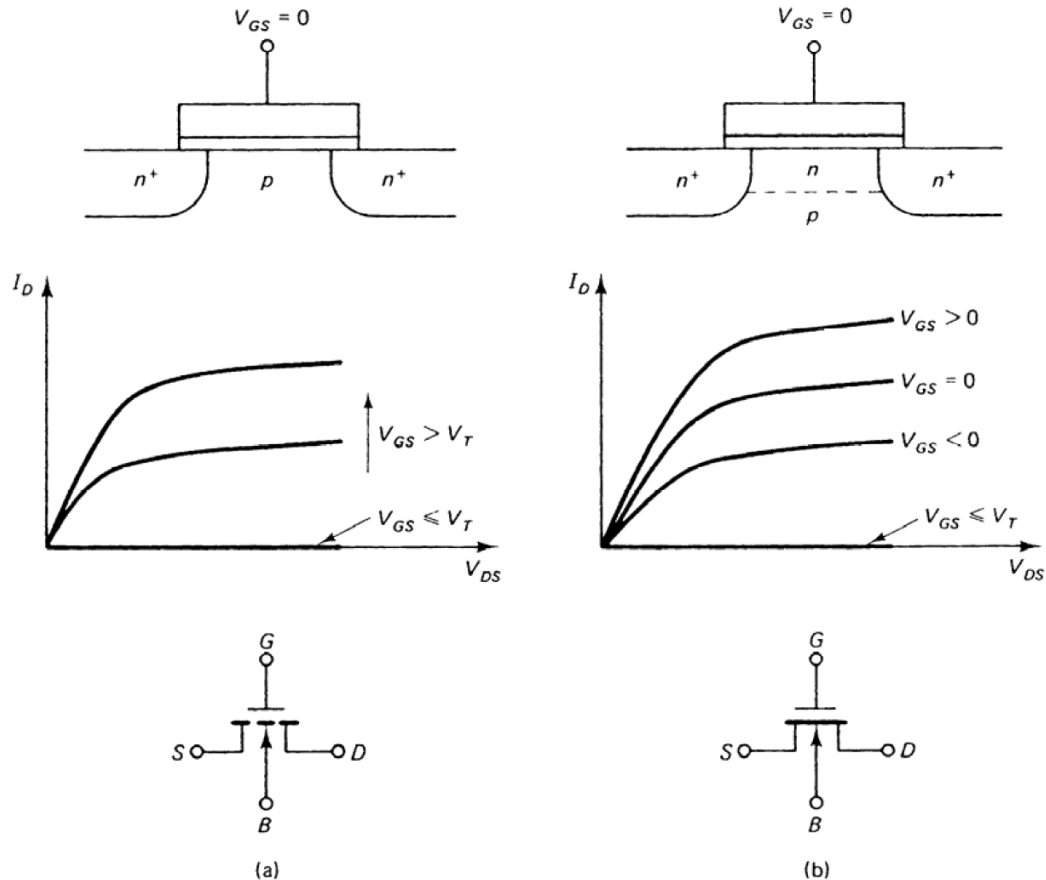
□ There are two different types of MOSFETs:

## □ Enhancement-mode MOSFETs

- This is also known as the “normally off” transistor. The transistor does not conduct any current when  $V_G=0$ .
- A positive  $V_G$  is required to turn on an n-MOSFET while a negative  $V_G$  is needed to turn on a p-MOSFET.

## □ Depletion-mode MOSFETs

- The transistor is normally on at  $V_G=0$ . A non-zero gate voltage is needed to turn off the transistor.
- A negative  $V_G$  is required to turn off an n-MOSFET while a positive  $V_G$  is needed to turn off a p-MOSFET.



Classes of MOSFET and typical  $I$ - $V$  characteristics: (a) n-channel enhancement and (b) n-channel depletion.

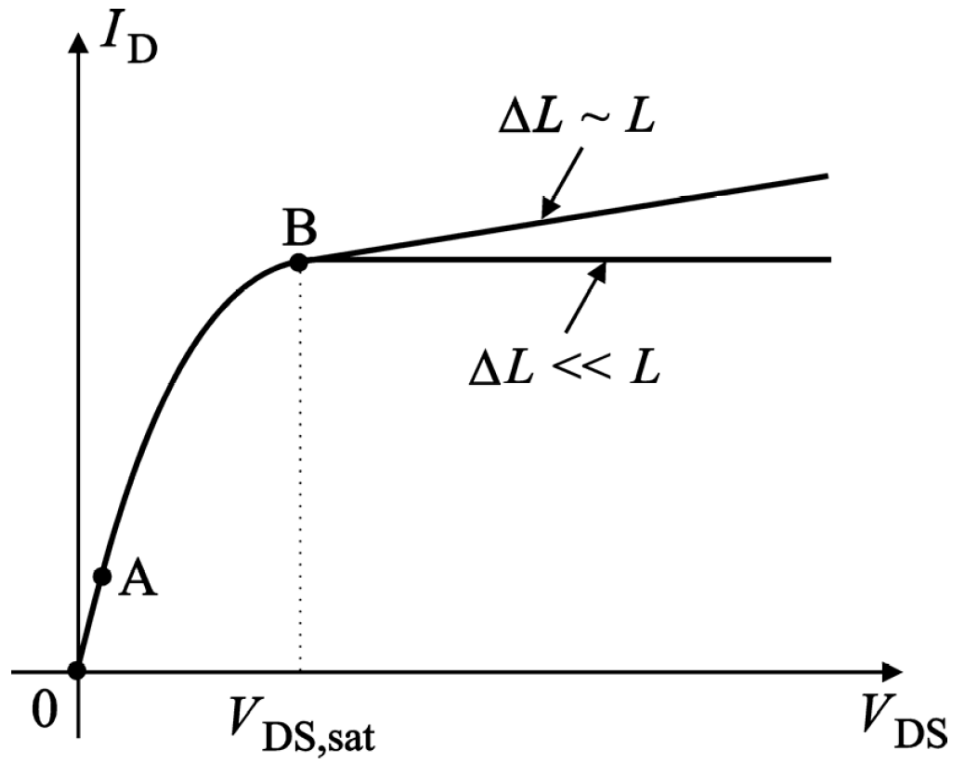
# MOSFET Operation

## Accumulation/Depletion

- ❑  $V_G < V_{TH}$ , where  $V_{TH}$  is the depletion to inversion transition voltage or threshold voltage.
- ❑ The semiconductor region under the gate contains either an excess (accumulation) or a deficit (depletion) of holes (majority carriers), and a very small number of minority carriers (electrons).
- ❑ The  $n^+$  source and drain regions are electrically isolated from one another, i.e. there is no conductive path for electrons.
- ❑ The transistor current  $I_D$  is therefore very small, consisting of only junction leakage currents. The transistor is said to be cut-off.

## Inversion

- ❑  $V_G > V_{TH}$ . An inversion layer, containing mobile electrons, is formed at the semiconductor region under the gate.
- ❑ This inversion layer (n-type) electrically connects the n+ drain and source regions, allowing a current to flow when  $V_{DS} > 0$ . The transistor is said to be turned-on.
- ❑ A larger  $V_G$  (relative to  $V_{TH}$ ) induces a larger concentration of electrons in the inversion layer. The channel conductance, and hence the drain current would be higher at larger  $(V_G - V_{TH})$ .
- ❑ For a given  $V_{DS}$ , the drain current can independently be controlled by the gate voltage. Hence, the term field effect transistor.



General variation of  $I_D$  with  $V_{DS}$  for a given  $V_{GS} > V_{TH}$ .

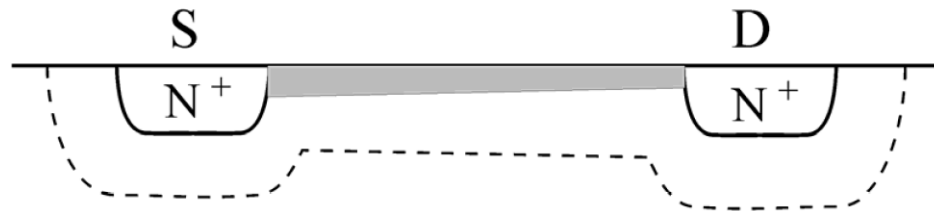


In inversion, how does the drain current  $I_D$  vary with the terminal voltages?

We will first examine how  $I_D$  varies with  $V_{DS}$  at a given  $V_{GS}$ . Consider the following:

$$V_{DS} = 0$$

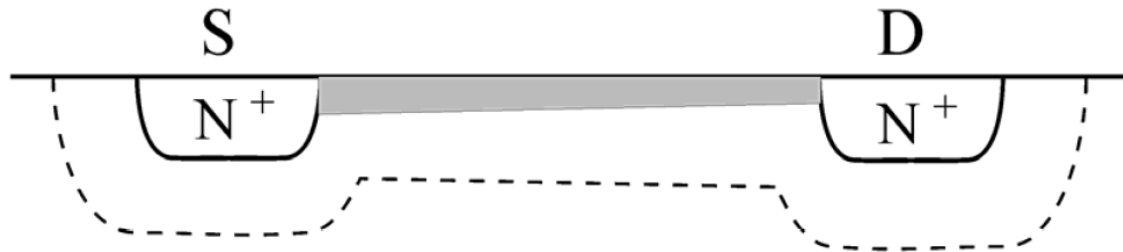
- ☐ The device is under thermal equilibrium and  $I_D = 0$ .
- ☐ The inversion layer is uniform along the length of the device.



$V_{DS} > 0$  (small positive voltage)

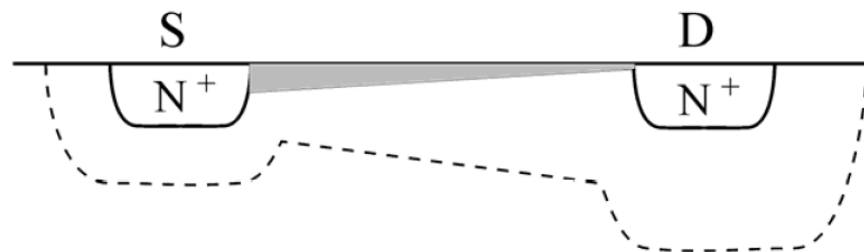
□  $I_D$  flows from D to S through an almost uniform channel. Hence the transistor behaves like a simple resistor, and  $I_D$  is proportional to  $V_{DS}$ .

□ The transistor is said to be in the linear region.



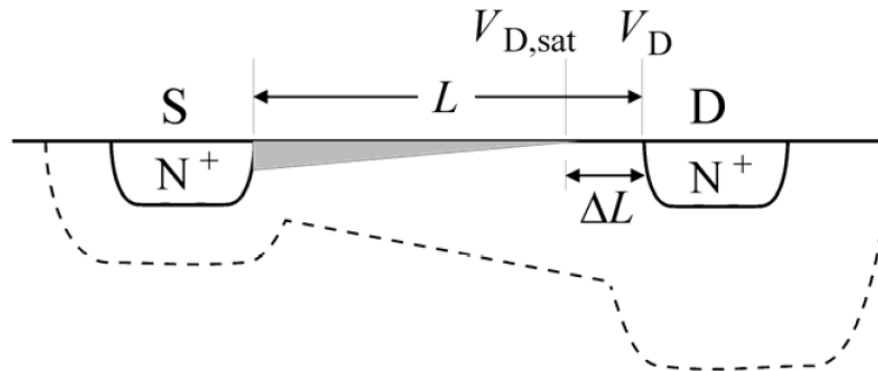
$V_{DS} > 0$  (higher positive voltage)

- ❑ The potential at the semiconductor surface becomes more positive at the drain end, compared to the source end.
- ❑ The drain end becomes less inverted than the source end since the effective potential difference between the semiconductor surface and the gate becomes smaller.
- ❑ The number of inversion charge carriers decreases toward the drain, as indicated by a ‘tapering’ of the inversion layer. This is accompanied by a widening of the depletion region near the drain end.
- ❑ Hence, the channel resistance increases, and the gradient of the current-voltage (I-V) curve decreases.



$V_{DS} > V_{GS} - V_{TH}$  (saturation region)

- ❑ With a sufficiently large  $V_{DS}$ , the channel completely vanishes at the drain end. The value of drain voltage ( $V_{DS,sat} = V_{GS} - V_{TH}$ ) at which this happens is known as the saturation or pinch-off drain voltage.
- ❑ As  $V_{DS}$  is increased further, the pinch-off point widens into a depletion region  $\Delta L$ . If the channel length  $L \gg \Delta L$ ,  $I_D$  remains essentially constant, since the source to pinch-off point will be almost identical in shape and have the same end point voltages ( $V_S = 0$  and  $V_{DS,sat}$ ) regardless of the value of  $V_{DS}$ . The rest of the voltage  $V_{DS} - V_{DS,sat}$  will be absorbed by  $\Delta L$ .



- Beyond pinch-off, the carriers flow by conduction through the channel until they reach the pinch-off point, where they will be swept across the depletion region by the high electric field.
- On the other hand, if  $\Delta L$  is comparable to  $L$ ,  $V_{DS,sat}$  will now appear across  $L - \Delta L \ll L$ . Hence, the current will increase with  $V_{DS}$  as shown.

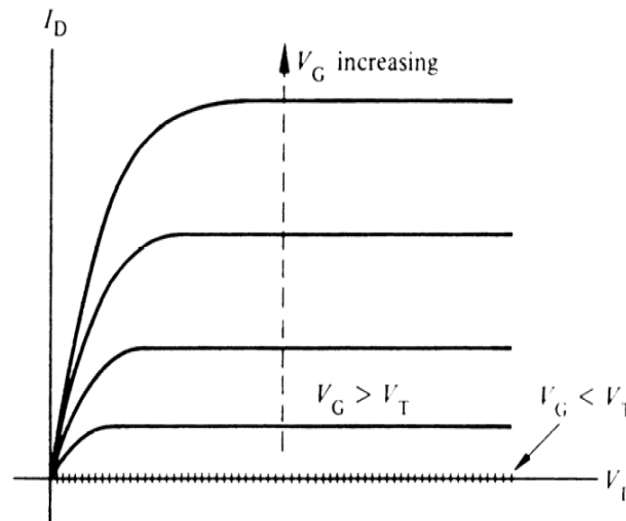
What about increasing  $V_{GS}$  at a given  $V_{DS}$  ?

$$V_{DS} < V_{DS,sat}$$

A higher  $V_{GS}$  would mean a larger channel conductance, since more electrons now reside near the surface. Correspondingly, a larger gradient is obtained as shown.

$$V_{DS} > V_{DS,sat}$$

$V_{DS,sat}$  is higher since  $V_{DS,sat} = V_{GS} - V_{TH}$ . This would then give rise to a larger drain saturation current  $I_{D,sat}$ .



The expression for the drain current in **the linear region** ( $V_{DS} \leq V_{GS} - V_{TH}$ ) can be approximated by an equation of the form:

$$I_D = \mu C_{ox} \frac{W}{L} \left[ (V_{GS} - V_{TH})V_{DS} - \frac{V_{DS}^2}{2} \right]$$

In **the saturation region** ( $V_{DS} \geq V_{GS} - V_{TH}$ ), the saturated drain current can be calculated as

$$I_{D,sat} = \mu C_{ox} \frac{W}{2L} (V_{GS} - V_{TH})^2$$

At the pinch-off point, both equations can be used for the drain current.